ISB 140/DS 110−INTRODUCTION TO STATISTICS

LECTURE NOTES

UNIT V: HYPOTHESIS TESTING

1. INTRODUCTION

- ✓ We have defined statistical inference as the process of making judgements about a population based on properties of a sample from the population.
- ✓ There are two branches of statistical inference such as **estimation** and **hypothesis testing**.
- ✓ In this session we shall dwell much on estimation and sampling distributions as well as Hypothesis testing which involves choosing between two opposing statements concerning a population.

2. ESTIMATION AND SAMPLING DISTRIBUTIONS

2.1. ESTIMATION

- ✓ Estimation involves approximating the value of an unknown parameter. Some commonly used statistics are given in the table below.

Measure	Parameter	Statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s
Proportion	p	$\hat{p}$

- ✓ Estimation can either be done by giving a point estimate or interval estimate (confidence interval).

(a) POINT ESTIMATE

- ➢ A point estimate is a single value (point) derived from a sample and is used to estimate the population parameters.
- ➢ In other words, a point estimate is the value of a sample statistic that is used to estimate a population parameter.

- For example, if we select a sample of 50 KMU student ages. We then compute the mean age of the 50 students and use the value of the mean as a **POINT ESTIMATOR** of unknown population mean.
- Note that a more informative approach is to present a range of values in which we expect the population parameter to fall.

(b) CONFIDENCE INTERVAL

- A confidence interval also called interval estimate present a range of values constructed from a sample data so that the population parameter is likely to fall within that range at a specified probability.
- The specified probability is known as the level of confidence.
- Suppose we let U and L be the Upper and the Lower confidence limit for the parameter θ based on a random sample $X_1, X_2, \dots, X_n$, then both L and U are functions of the sample, and we can write the interval estimate of θ as

$$P(L \leq \theta \leq U) = 1 - \alpha.$$

- The interpretation of this interval estimate is that we are $100\ (1 - \alpha)$ % certain that the actual parameter θ falls in the interval (L, U).
- The quantity $1 - \alpha$ is known as the confidence level, and the interval (U, L) is called $100(1 - \alpha)\%$ confidence interval for θ.

2.2. SAMPLING DISTRIBUTIONS

- ✓ To evaluate the reliability of our inferences, we will need to know the probability distribution for the statistic we are using.

Definition (***Sampling Distribution***)

- ✓ Sampling distribution is the probability distribution of a sample statistic.
- ✓ In other words, sampling distribution is the distribution of results if you actually selected all possible samples.

Definition (sample size)

- ✓ A sample is a set of observable random variables, $X_1, \dots, X_n$. The number n is called the sample size.

Definition (identically distributed)

- ✓ A random sample of size n from a population is a set of n independent and identically distributed (iid) observable random variables $X_1, \dots, X_n$.

- ✓ We call the random variables $X_1, \dots, X_n$ identically distributed if every X_i has the same probability distribution.

NOTE

(i) In a sample (not a random sample), X_i need not be independent or identically distributed.

(ii) Random sampling helps us to control systematic biases.

(iii) For a finite population, one can successively number the elements of the population and then select a random sample with the help of a table of random digits.

(iv) One of the simplest ways to select a random sample of finite size is to use a table of random numbers

Definition (statistic)

- ✓ A function T of observable random variables $X_1, \dots, X_n$ that does not depend on any unknown parameters is called a statistic.
- ✓ The sample mean $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ is a function of $x_1, \dots, x_n$.

EXAMPLE

Let the population consist of the numbers $\{1,2,3,4,5\}$. Consider all possible samples consisting of three numbers randomly chosen without replacement from this population. Obtain the distribution of the sample mean.

SOLUTION

Disregarding the order, it is clear that there are $\binom{5}{3} = 10$ equally likely possible samples of size 3.

They are (1,2,3), $(1,2,4), (1,2,5), (1,3,4), (1,3,5), (1,4,5), (2,3,4), (2,3,5), (2,4,5)$, and $(3,4,5)$. Calculating the sample mean, $\bar{x}$, for each of the samples, we will get the sampling distribution of $\overline{x}$ as:

$\bar{x}$	$\frac{2}{1}$	$\frac{7}{3}$	$\frac{8}{3}$	$\frac{3}{1}$	$\frac{10}{3}$	$\frac{11}{3}$	$\frac{4}{1}$
$p(\bar{x})$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{1}{10}$

For example, in the table, $P(\bar{x} = 8/3) = 2/10$ because the two samples $(1,2,5)$ and $(1,3,4)$ both give an $\bar{x} = 8/3$, which is an estimate of the population mean, μ.

In general, sampling distributions are theoretical distributions that consist of possibly an infinite number of sample statistics taken from an infinite number of randomly selected samples of a fixed sample size. Every sample statistic has a sampling distribution.

Theorem (*): Let $X_1, \dots, X_n$ be a random sample of size n from a population with mean μ and variance σ^2. Then $E(\bar{x}) = \mu$ and $\text{Var}(\bar{x}) = \sigma^2/n$.

Proof. The mean and variance of $\bar{x}$ are given by,

$$E(\bar{x}) = E\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(x_i)$$
$$= \frac{1}{n}\sum_{i=1}^{n} \mu = \frac{1}{n}n\mu = \mu$$

and

$$\text{Var}(\bar{x}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)$$
$$= \frac{1}{n^2}\sum_{i=1}^{n} \text{Var}(x_i)\big(\text{ because } x_i's \text{ are independent and } \text{Var}(ax_i) = a^2\text{Var}(x_i)\big)$$
$$= \frac{1}{n^2}n\sigma^2 = \frac{\sigma^2}{n}$$

Definition (Finite populations)

- ✓ Let $\{c_1, c_2, \dots, c_N\}$ be a finite population. Then the population mean $\mu = \frac{1}{N}\sum_{i=1}^{N} c_i$ and the population variance $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N} (c_i - \mu)^2$.
- ✓ The following theorem for the sample mean and variance is stated without proof.

Theorem: If $X_1, \dots, X_n$ is a sample of size n (chosen without replacement) from a population $\{c_1, c_2, \dots, c_N\}$, then:

$$E(\bar{x}) = \mu$$
and
$$\text{Var}(\bar{x}) = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)$$

NOTE

(i) The sample in the theorem is not a random sample and $X_i's$ are not iid random variables.

(ii) The factor $(N-n)(N-1)$ in the foregoing theorem is often called the finite population correction factor.

(iii) The sampling without replacement causes dependence among the $X_i s$.

(iv) If the sample size n is small relative to the population size N, the population correction factor is approximately 1.

(v) We will not use the finite population correlation factor in the derivation of a sampling distribution, unless it is absolutely necessary.

EXAMPLE

Obtain the mean and variance of $\bar{x}$ using the information from the previous Example

SOLUTION

First note that for the population in the previous Example, the population mean is

$$\mu = \frac{1}{\mathrm{N}} \sum_{\mathrm{i}=1}^{\mathrm{N}} \mathrm{c_i} = 3$$

and the population variance is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (\mathrm{c_i} - \mu)^2 = 2$$

Applying the probability distribution of $\bar{x}$ given in the Example above, we obtain:

$$E(\bar{x}) = 2\left(\frac{1}{10}\right) + \frac{7}{3}\left(\frac{1}{10}\right) + \frac{8}{3}\left(\frac{2}{10}\right) + 3\left(\frac{2}{10}\right) + \frac{10}{3}\left(\frac{2}{10}\right) + \frac{11}{3}\left(\frac{1}{10}\right) + 4\left(\frac{1}{10}\right)$$
$$= 3$$

EXAMPLE

Consider a small population of board members of an organization. Let x = number of children that each board member has.

Board member	1	2	3	4	5	6	7	8
Number of children	2	2	0	0	2	2	0	3

i.e. board member 1 has 2 children, board member 2 has 2 children, etc…

The average number of children for this population is

$$\mu = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{2+2+0+0+2+2+0+3}{8} = \frac{11}{8} = 1.38$$

To estimate μ we pick simple random samples of size 2. How many samples exist?

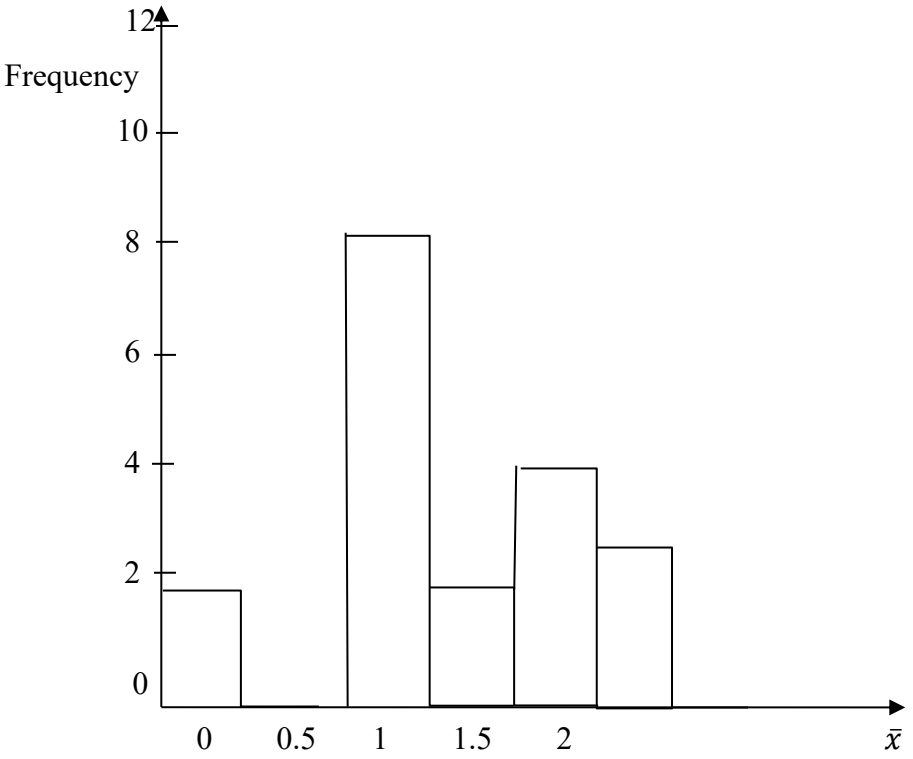
$$\binom{8}{2} = \frac{8!}{2!\,6!} = 28$$

We can list them and calculate the mean of each of them as follows:

	Sample	$\bar{x}$	
1	(1, 2)	2	i.e. mean for board members 1 and 2 $= \frac{2+2}{2} = 2$
2	(1, 3)	1	i.e. mean for board members 1 and 3 $= \frac{2+0}{2} = 1$
3	(1, 4)	1	i.e. mean for board members 1 and 4 $= \frac{2+0}{2} = 1$
4	(1, 5)	2	i.e. mean for board members 1 and 5 $= \frac{2+2}{2} = 2$
	⋮	⋮	
28	(7, 8)	1.5	i.e. mean for board members 7 and 8 $= \frac{0+3}{2} = 1.5$

(You can complete the whole table as an exercise)

A histogram of the $\bar{x}$ values is given as



✓ Observations:

(i) The histogram tends to centered around the mean μ.

(ii) The histogram resembles a bell-shaped distribution.

(iii) Generally, the sampling distribution of $\bar{x}$ resembles the normal distribution (actually the distribution will look more normal as the number of samples increase).

EXAMPLE

Let $X_1, \ldots, X_n$ be a random sample from a population with mean μ and variance σ^2.

Consider the sample variance:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

Show that $E(s^2) = \sigma^2$.

SOLUTION

It can be shown that

$$\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}{n-1}$$

Hence,

$$E(S^2) = E\left(\frac{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2}{n-1}\right) = \frac{1}{n-1}\sum_{i=1}^{n} E(x_i^2) - \frac{n}{n-1}E(\bar{x}^2).$$

Using the fact that $\mathrm{E}(x^2) = \mathrm{Var}(x) + \mu^2$ and Theorem (*), we have.

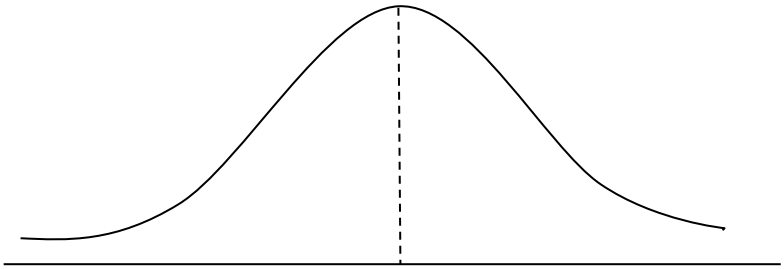
$$\begin{aligned} E(S^2) &= \frac{1}{n-1}n(\sigma^2 + \mu^2) - \frac{n}{n-1}\left(\frac{\sigma^2}{n} + \mu^2\right) \\ &= \left(\frac{n}{n-1} - \frac{1}{n-1}\right)\sigma^2 + \left(\frac{n}{n-1} - \frac{n}{n-1}\right)\mu^2 \\ &= \sigma^2 \end{aligned}$$

A. NORMAL DISTRIBUTION

- ✓ A normal distribution is very important distribution in statistics. A normal distribution is characterized by its mean μ and variance σ^2.
- ✓ If a random variable X is normally distributed with mean μ and variance σ^2, then we write $X \sim N(\mu, \sigma^2)$.
- ✓ The graph of a normal distribution is called a normal curve.

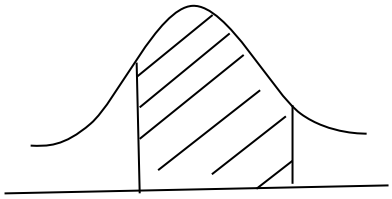
PROPERTIES OF A NORMAL CURVE

1. The normal curve is bell shaped and has a peak over the mean μ.



μ

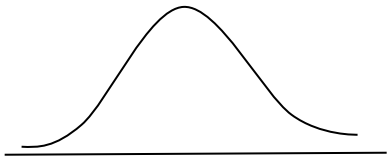
1. It is symmetrical about a vertical line through μ.
2. The probability that X is between a and b is given by the area under the curve i.e.



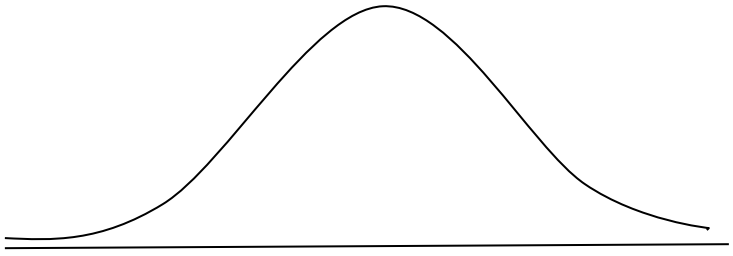
a b

$$P(a < X < b) = Area\ below\ the\ curve\ between\ a\ and\ b$$
$$= P(a < X \leq b)$$
$$= P(a \leq X \leq b)$$
$$= P(a \leq X < b)$$

3. The parameters σ^2 controls the spread of the curve e.g.



$\sigma^2 = 5$



$\sigma^2 = 10$

i.e. the second curve is more variable so it will have a bigger variance.

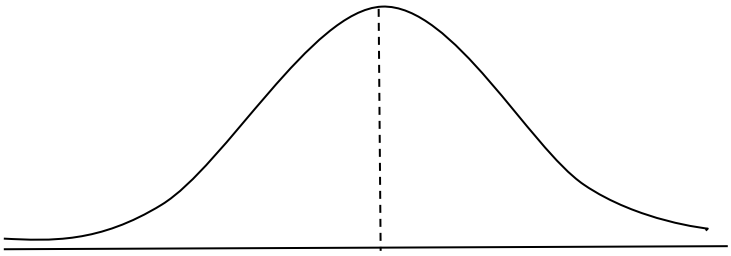
B. STANDARD NORMAL

✓ If $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma}$ is a standard normal distribution and $Z \sim N(0,1)$

i.e. Z is a normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 1$.

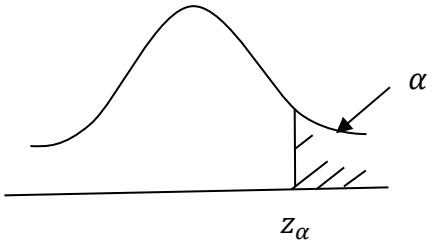
NOTE

1. The standard normal curve is centered on 0.



0

2. The total area under the curve is 1 and $P(Z < 0) = 0.5$, $P(Z > 0) = 0.5$.
3. If $P(Z > z_\alpha) = \alpha$ then



4. Instead of integrating, probabilities associated with the standard normal distribution is obtained from the standard normal table.
5. Probabilities associated with any normal distribution are also obtained by first standardizing (converting to a standard normal) and then using the standard normal tables.

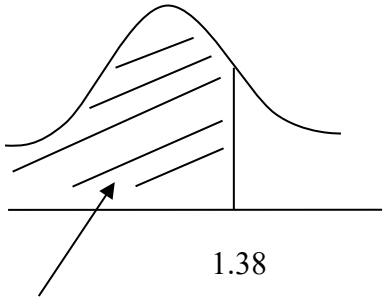
✓ Below are some examples on how this is done.

EXAMPLES

1. If $Z \sim N(0,1)$, find
   (a) $P(Z < 1.38)$
   (b) $P(Z > 1.28)$
   (c) $P(1.12 < Z < 1.41)$
   (d) $P(-2.02 < Z < 1.74)$
   (e) $z_{0.025}$

SOLUTION

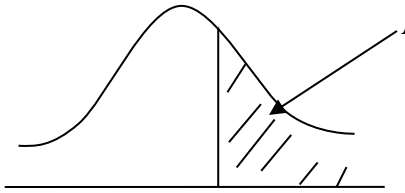
(a)



Area below 1.38 (refer to the standard normal (Z) table given)

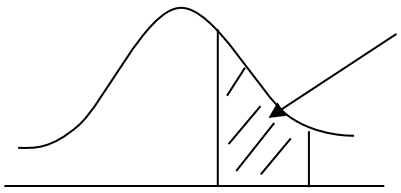
$$P(Z < 1.38) = 0.9162$$

(b)



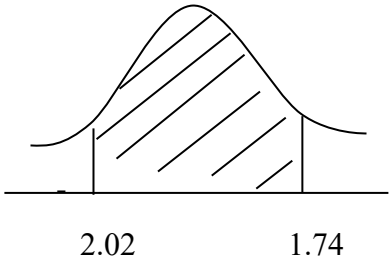
$$\begin{aligned} P(Z > 1.28) &= 1 - P(Z < 1.28) \\ &= 1 - 0.8997 \\ &= 0.1003 \end{aligned}$$

(c)



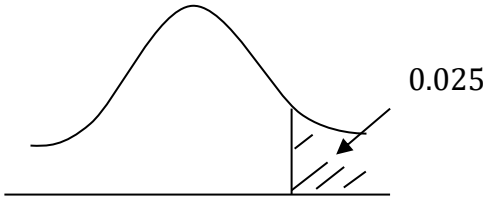
$$\begin{aligned} P(1.12 < Z < 1.41) &= P(Z < 1.41) - P(Z < 1.12) \\ &= 0.9207 - 0.8686 \\ &= 0.0521 \end{aligned}$$

(d)



$$
\begin{aligned}
P(-2.02 < Z < 1.74) &= P(Z < 1.74) - P(Z < -2.02) \\
&= 0.9591 - P(Z > 2.02) \ \ (by\ symmetry) \\
&= 0.9591 - (1 - P(Z < 2.02)) \\
&= 0.9591 - (1 - 0.9783) \\
&= 0.9374
\end{aligned}
$$

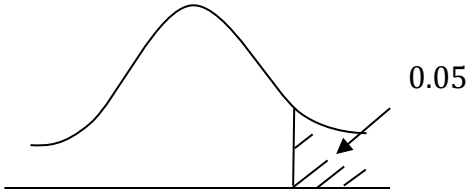
(e)



$z_{0.025}$

$$
\begin{aligned}
z_{0.025} & \\
P(Z > z_{0.025}) &= 0.025 \\
P(Z < z_{0.025}) &= 1 - 0.025 \\
P(Z < z_{0.025}) &= 0.975 \\
z_{0.025} &= 1.96\ (z\ value\ giving\ a\ lower\ 0.975\ probability)
\end{aligned}
$$

(f)



$z_{0.05}$

$$
\begin{aligned}
z_{0.05} & \\
P(Z > z_{0.05}) &= 0.05 \\
P(Z < z_{0.05}) &= 1 - 0.05 \\
P(Z < z_{0.05}) &= 0.95 \\
z_{0.05} &= 1.645\ (z\ value\ giving\ a\ lower\ 0.95\ probability)
\end{aligned}
$$

NOTE

1. Parts (e) and (f) can be directly obtained using the last two rows of the standard normal tables, just note that z_α in our notation refers to the z value that leaves an upper tail area of α. In the normal table z_γ refers to a z value that leaves a lower tail area of γ. Thus $z_\alpha = z_{1-\gamma}$.
2. The heights of boys in a certain class follow a normal distribution with mean 150.3cm and standard deviation 5cm. Find the probability that a boy picked at random from this class has a height
   (a) less than 153cm.

(b) more than 144cm.

(c) between 147cm and 149.5cm.

SOLUTION

(a) $X \sim N(150.3, 5^2)$ $\quad i.e. \quad \mu = 150.3,\ \sigma = 5$

$$P(X < 153) = P\left(\frac{X-\mu}{\sigma} < \frac{153-150.3}{5}\right) \ i.e. standardize\ to\ use\ Z\ tables$$
$$= P(Z < 0.54)$$
$$= 0.7054$$

(b)

$$P(X > 144) = P\left(\frac{X-\mu}{\sigma} > \frac{144-150.3}{5}\right)$$
$$= P(Z > -1.26)$$
$$= P(Z < 1.26)$$
$$= 0.8962$$

(c)

$$P(147 < X < 149.5) = P\left(\frac{147-150.3}{5} < \frac{X-\mu}{\sigma} < \frac{149.5-150.3}{5}\right)$$
$$= P(-0.66 < Z < -0.16)$$
$$= P(0.16 < Z < 0.66)$$
$$= P(Z < 0.66) - P(Z < 0.16)$$
$$= 0.7454 - 0.5636$$
$$= 0.1818$$

C. SAMPLING DISTRIBUTION OF $\bar{X}$ (NORMAL POPULATION)

- ✓ If $X_1, X_2, \ldots, X_n$ is a random sample of size n taken from a normal distribution with mean μ and variance σ^2 such that $X \sim N(\mu, \sigma^2)$, then the distribution of $\bar{X}$ is also normal and $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ where $\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$.

  NOTE

  1. We use capital letters to represent random variables i.e. $\bar{X}$ is a random variable, $\bar{x}$ is an observed value of $\bar{X}$.
  2. $\frac{\sigma}{\sqrt{n}}$ is called the standard error of the mean.
  3. The mean of $\bar{X}$ is μ i.e. $E(\bar{X}) = \mu$ or $\mu_{\bar{X}} = \mu$ and the variance of $\bar{X}$ is $\frac{\sigma^2}{n}$ or $Var(\bar{X}) = \frac{\sigma^2}{n}$ or $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$ i.e. $Z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}$ is a standard normal.

EXAMPLE

A normal distribution has a mean of 40 and a standard deviation of 4. If 25 items are drawn at random, find the probability that their mean is

(a) 41.4 or more

(b) between 38.7 and 40.7

(c) less than 39.5

SOLUTION

(a)

$$X \sim N(40, 4^2) \ , \qquad n = 25$$

$$\bar{X} \sim N\left(40, \frac{4^2}{25}\right) \ i.e. \ \sigma_{\bar{X}} = \frac{4}{5}$$

$$\begin{aligned} P(\bar{X} \geq 41.4) &= P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{41.4 - 40}{\frac{4}{5}}\right) \\ &= P(Z > 1.75) \\ &= 1 - P(Z < 1.75) \\ &= 1 - 0.9599 \\ &= 0.0401 \end{aligned}$$

(b)

$$\begin{aligned} P(38.7 < \bar{X} < 40.7) &= P\left(\frac{38.7 - 40}{\frac{4}{5}} < \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{40.7 - 40}{\frac{4}{5}}\right) \\ &= P(-1.625 < Z < 0.875) \\ &= P(Z < 0.875) - P(Z < -1.625) \\ &= 0.8092 - P(Z > 1.625) \\ &= 0.8092 - (1 - P(Z < 1.625)) \\ &= 0.8092 - (1 - 0.9479) \\ &= 0.7571 \end{aligned}$$

NOTE

1. To find $P(Z < 0.875)$ we go halfway between 0.87 and 0.88, we get the average of the probabilities for these two i.e. $\frac{0.8078+0.8106}{2} = 0.8092$.
2. Similarly, to find $P(Z < 1.625)$ we find the probability which is halfway between 1.62 and 1.63 i.e. $\frac{0.9474+0.9484}{2} = 0.9479$

(c)

$$
\begin{aligned}
P(\bar{X} < 39.5) &= P\left(\frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}} < \frac{39.5-40}{\frac{4}{5}}\right) \\
&= P(Z < -0.625) \\
&= P(Z > 0.625) \\
&= 1 - P(Z < 0.625) \\
&= 1 - 0.73405 \\
&= 0.26595 \\
&= 0.2660
\end{aligned}
$$

A. CENTRAL LIMIT THEOREM

- ✓ If $X_1, X_2, \ldots, X_n$ is a random sample of size n taken from a any distribution with mean μ and variance σ^2 then, for large n, the distribution of the sample mean $\bar{x}$ is approximately normal and $\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ where $\bar{x} = \frac{1}{n}(x_1 + x_2 + \cdots + x_n)$.

NOTE

1. Provided n is large enough, the shape of the sampling distribution of $\bar{X}$ is approximately normal i.e.

$$\frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} \xrightarrow{d} Z \sim N(0,1)$$

Read as "converges in distribution to $N(0,1)$". The more the underlying population departs from normality, the larger the value of n necessary to ensure normality.

2. How large should n be? Generally, 30 or more.

2.3. ONE SAMPLE ESTIMATION (BASED ON ONE SAMPLE)

- ✓ Suppose we want to estimate the average salary of all civil servants in Zambia, we consider the following procedure:

  (1) We select a random sample of n civil servants.

  (2) Calculate $\bar{x}$ based on our sample

  (3) Use $\bar{x}$ to estimate μ = average salary of all civil servants

- ✓ The sample mean $\bar{x}$ is an example of a point estimate for the population mean μ. It is obvious that $\bar{x}$ is (almost) never equal to μ.

- ✓ To solve this problem, we give a range of possible values of μ or construct a confidence interval, and this process is called interval estimation.

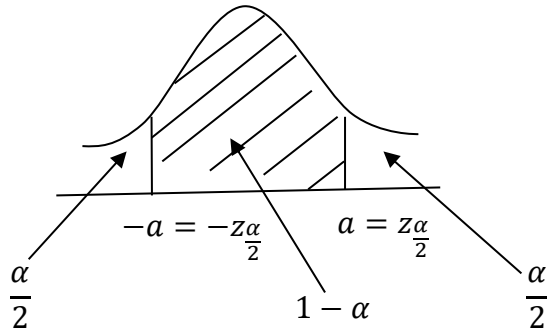
A. ESTIMATING CONFIDENCE INTERVALS FOR A POPULATION MEAN μ BASED ON ONE SAMPLE

- ✓ To construct a confidence interval for the population, mean μ, we consider three cases:

Case 1: μ is unknown and Population variance σ^2 is known, X is normally distributed

- ✓ Case 1 is obtained when the population mean μ is un known and the population variance σ^2 is known, that is, $X_1, X_2, \ldots, X_n$ is a random sample from a population random variable X and $X \sim N(\mu, \sigma^2)$, where μ is unknown and σ^2 is known. Let

$$P\left(-a < \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} < a\right) = 1-\alpha \ \ (0<\alpha<1)$$



Then

$$P\left(-z_{\frac{\alpha}{2}} < \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} < z_{\frac{\alpha}{2}}\right) = 1-\alpha$$

$$P\left(-z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} < \bar{x}-\mu < z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha \ \left(multiply\ by\ \frac{\sigma}{\sqrt{n}}\right)$$

$$P\left(z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} > \mu-\bar{x} > -z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha \ (multiply\ by -1)$$

$$P\left(-z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} < \mu-\bar{x} < z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$$

$$P\left(\bar{X}-z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x}+z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$$

i.e. the probability that μ is in the interval $\left(\bar{x}-z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\ ,\ \bar{x}+z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right)$ is $1-\alpha$.

- ✓ When a particular sample is picked from the population, we have the interval

$$\left(\bar{x} - z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\ , \qquad \bar{x} + z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right)$$

- ✓ Thus, if $\bar{x}$ is the mean of a random sample of size n taken from a normal population with known variance σ^2, then a $100(1-\alpha)\%$ confidence interval for μ is given by

$$C.I: \bar{x} \pm z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} = \left(\bar{x} - z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}}\right).$$

- ✓ The margin of error for the estimate of μ, denoted by E or e, is the quantity that is subtracted from and added to the value of $\bar{x}$ to obtain a confidence interval for μ.
- ✓ Thus, the margin of error is given by $E = z\sigma_{\bar{x}}\ or\ e = z\sigma_{\bar{x}}$, where $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, implying that $E = z\sigma_{\bar{x}} = z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} = e$

NOTE

1. Suppose $\alpha = 0.05$ then we have a 95% confidence interval. What does a 95% confidence interval mean? It means if we were to take 100 samples of size n from the population and construct 100 confidence intervals then approximately 95 of them will contain μ .
2. If $\bar{x}$ is used to estimate μ, we can be $100(1-\alpha)\%$ confident that the error will not exceed a specified value e when the sample size is

$$n = \left(\frac{\sigma z_{\frac{\alpha}{2}}}{e}\right)^2$$

i.e.

$$\begin{aligned} e &= z_{\frac{\alpha}{2}}\frac{\sigma}{\sqrt{n}} \\ e^2 &= \frac{z_{\frac{\alpha}{2}}^2\sigma^2}{n} \\ n &= \frac{z_{\frac{\alpha}{2}}^2\sigma^2}{e^2} = \left(\frac{\sigma z_{\frac{\alpha}{2}}}{e}\right)^2 \end{aligned}$$

EXAMPLE

After a rainy night, 12 worms surfaced in the garden. The average length of the 12 worms is calculated to be 10.39cm. Assuming that the sample came from a normal population with variance 4.

(a) find a 95% confidence interval for the mean length of all worms in the garden.

(b) find a 90% confidence interval for the mean length of all worms in the garden.

(c) how large a sample is required if we want to be 95% confident that our estimate of μ is off by less than 0.5cm.

SOLUTION

(a)

$$X \sim N(\mu, 4)$$

$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \ , \alpha = 0.05$$

$$10.39 \pm z_{0.025} \frac{2}{\sqrt{12}}$$

$$10.39 \pm 1.96 \left(\frac{2}{\sqrt{12}}\right)$$

$$10.39 \pm 1.1316$$

$$(9.26, 11.52)$$

(b)

$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \ , \alpha = 0.1$$

$$10.39 \pm z_{0.05} \frac{2}{\sqrt{12}}$$

$$10.39 \pm 1.645 \left(\frac{2}{\sqrt{12}}\right)$$

$$10.39 \pm 0.94974$$

$$(9.44, 11.34)$$

(c)

$$n = \left(\frac{\sigma z_{\frac{\alpha}{2}}}{e}\right)^2 \ , e = 0.5$$

$$n = \left(\frac{2z_{0.025}}{0.5}\right)^2$$

$$n = \left(\frac{2(1.96)}{0.5}\right)^2$$

$$n = 61.46 \ (we \ usually \ "round \ up")$$

$$n = 62$$

Case 2: X is Normally distributed and population variance σ^2 is unknown

- ✓ In this case, we assume $X_1, X_2, \dots, X_n$ is a random sample from a population random variable X and $X \sim N(\mu, \sigma^2)$, where μ and σ^2 are unknown.
- ✓ If we pick a random sample from a normal population with an unknown variance σ^2, we estimate σ^2 by s^2, where

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$= \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2\right]$$

- ✓ If we replace σ by s then the random variable

$$T_{obs} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

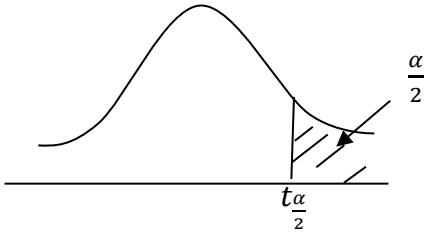
follows a student's t − distribution with $n - 1$ degrees of freedom i.e.

$$\frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \sim t_{n-1}$$

- ✓ Thus, if $\bar{x}$ and s are the mean and standard deviation of a random sample of size n ($n < 30$) taken from a normal population with an unknown variance σ^2, then a $100(1-\alpha)\%$ confidence interval for μ is given by

$$C.I: \bar{x} \pm t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}} = \left(x - t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}}, x + t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}}\right)$$

where $t_{\frac{\alpha}{2},n-1}$ is a t – value with $n - 1$ degrees of freedom, leaving an upper tail area of $\frac{\alpha}{2}$ i.e.



NOTE

1. The t − distribution is symmetric about 0 just like the standard normal distribution.

2. Statistical tables for the t − distribution that we use in this course give quantiles of the upper tail probability. To read the tables we pick the appropriate degrees of freedom (df) from the first column and the appropriate tail probability from the first row.
3. As the degrees of freedom approaches infinity, the t − distribution approaches the standard normal distribution.

EXAMPLE

The contents of 7 similar containers of sulphuric acid are 9.8, 10.2, 10.4, 9.8, 10.0, 10.2 and 9.6 litres. Find a (a) 95% (b) 99% confidence intervals for the mean of all such containers, assuming an approximate normal distribution.

SOLUTION

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{9.8 + 10.2 + \cdots + 9.6}{7} = \frac{70}{7} = 10$$

$$\begin{aligned} s^2 &= \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n}\right] \\ &= \frac{1}{6}\left[700.48 - \frac{70^2}{7}\right] \\ &= 0.08 \end{aligned}$$

Note that the mean $\bar{x}$ and standard deviation s can be obtained directly from your scientific calculator (by the way you need a scientific calculator in this course). Simply put your calculator in statistics mode (check the manual on how to do this), enter the observations and press mean $\bar{x}$ and standard deviation s and you will find $\bar{x} = 10$, $s = 0.2828427$.

(a)

$$\bar{x} \pm t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}} \ , \alpha = 0.05$$

$$10 \pm t_{0.025,6}\sqrt{\frac{0.08}{7}}$$

$$10 \pm 2.447\sqrt{\frac{0.08}{7}}$$

$$(9.74, 10.26) \; or \; 9.74 < \mu < 10.26$$

(b)

$$\bar{x} \pm t_{\frac{\alpha}{2},n-1}\frac{s}{\sqrt{n}} \ , \alpha = 0.01$$

$$10 \pm t_{0.005,6}\sqrt{\frac{0.08}{7}}$$

$$10 \pm 3.707\sqrt{\frac{0.08}{7}}$$

$$(9.60, 10.40)$$

Case 3: X is not Normally distribution and σ^2 is unknown

- ✓ If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from an unknown distribution with an unknown variance σ^2, then provided $n \geq 30$ (large), we can replace σ by *s* and a $100(1-\alpha)\%$ confidence interval for μ is given by

$$C.I: \bar{x} \pm z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}} = \left(\bar{x} - z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}}\frac{s}{\sqrt{n}}\right)$$

- ✓ This is often referred to as a **LARGE – SAMPLE CONFIDENCE INTERVAL**.

NOTE

1. Cases 1 and 2 are usually referred to as small – sample confidence intervals. We need the normal assumption for these two cases, the only distinction between them is that σ^2 is unknown in one case (case 2).
2. Case 3 is based on the central limit theorem.

EXAMPLE

The heights of a random sample of 50 college students showed a mean of 174.5cm and a standard deviation of 6.9cm. Construct a (a) 95% and (b) 98% confidence interval for the mean height of all college students.

SOLUTION

$n = 50, \quad \bar{x} = 174.5,$

$s = 6.9 \ (sample\ standard\ deviation\ since\ it\ is\ based\ on\ the\ sample)$

(a)

$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \ , \alpha = 0.05$$

$$174.5 \pm z_{0.025} \frac{6.9}{\sqrt{50}}$$

$$174.5 \pm 1.96 \left(\frac{6.9}{\sqrt{50}}\right)$$

$$(172.59, 176.41)$$

(b)

$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \ , \alpha = 0.02$$

$$174.5 \pm z_{0.01} \frac{6.9}{\sqrt{50}}$$

$$174.5 \pm 2.326 \left(\frac{6.9}{\sqrt{50}}\right)$$

$$(172.23, 176.77)$$

B. ESTIMATING CONFIDENCE INTERVALS FOR A POPULATION PROPORTION BASED ON ONE SAMPLE

- ✓ Considering a binomial distribution with parameter p, we derive the population proportion with parameter p.
- ✓ Let p be a proportion of elements in a population possessing a certain characteristic of interest.
- ✓ The parameter p is called a population proportion. We estimate the population proportion p by its sample proportion $\hat{p}$ given by

$$\hat{p} = \frac{x}{n}$$

where x = number of elements in the sample possessing the characteristic of interest (or number of successes) and n = number of elements in the sample.

NOTE

1. $\hat{p} = \frac{x}{n}$ is a point estimate of p and $\hat{P} = \frac{X}{n}$ is a point estimator of p.
2. There are only two possible outcomes, an element has the characteristic of interest or it doesn't. Usually having a characteristic of interest is taken as a success and not having the characteristic as a failure.

- ✓ It can be shown that $\hat{P}$ is approximately normal with mean $E(\hat{P}) = \mu_{\hat{P}} = p$ and variance $Var(\hat{P}) = \sigma_{\hat{P}}^2 = \frac{p(1-p)}{n}$ recall that $X \sim BIN(n, p)$ so,

$$E(X) = np \text{ and } Var(X) = np(1-p).$$

Implying that $E(\hat{P}) = E\left(\frac{X}{n}\right) = \frac{E(X)}{n} = \frac{np}{n} = p$ and

$$Var(\hat{P}) = Var\left(\frac{X}{n}\right) = \frac{Var(X)}{n^2} = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$$

- ✓ Thus, if $\hat{p}$ is the proportion of successes in a random sample of size n, an approximate $100(1-\alpha)\%$ confidence interval for a population proportion p is given by

$$\hat{p} \pm z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

when the population proportion p is known.

NOTE

1. The normal approximation is valid provided $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$ i.e. n is large and $\hat{p}$ is not too close to 0 or 1.
2. If $\hat{p}$ is used as an estimate of p, we can be $100(1-\alpha)\%$ confident that the error will be less than a given value e when the sample size is approximately

$$n = \frac{z_{\frac{\alpha}{2}}^2 \hat{p}(1-\hat{p})}{e^2}$$

$$\text{i. e. obtained by solving} \quad e = z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad \text{for } n$$

3. If $\hat{p}$ is used as an estimate of p, we can be at least $100(1-\alpha)\%$ confident that the error will be less than a given value e when the sample size is approximately

$$n = \frac{z_{\frac{\alpha}{2}}^2}{4e^2}$$

i.e. replace $\hat{p}(1-\hat{p})$ by its maximum value $\frac{1}{4}\left(\hat{p} = \frac{1}{2}\right)$.

EXAMPLE

In a random sample of 500 families in Lusaka it was found that 340 watch sports

(a) Find a 95% confidence interval for the actual proportion of families in Lusaka who watch sports.

(b) How large a sample is required if we want to be

(i) 95% confident that our estimate of p is within 0.02 units of the true value?

(ii) at least 95% confident that our estimate of p is within 0.02 units of the true value?

SOLUTION

(a)

$$n = 500, \ x = 340, \qquad \alpha = 0.05$$

$$\hat{p} \pm z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\frac{340}{500} \pm z_{0.025}\sqrt{\frac{\frac{340}{500}\left(1-\frac{340}{500}\right)}{500}}$$

$$0.68 \pm 1.96\sqrt{\frac{0.68(0.32)}{500}}$$

$$0.68 \pm 0.0409$$

$$0.639 < p < 0.721$$

(b)

(i) $e = 0.02$

$$\begin{aligned} n &= \frac{z_{\frac{\alpha}{2}}^2 \hat{p}(1-\hat{p})}{e^2} \\ &= \frac{z_{0.025}^2(0.68)(1-0.68)}{0.02^2} \\ &= \frac{1.96^2(0.68)(0.32)}{0.02^2} \\ &= 2089.8 \\ n &= 2090 \ (remember\ to\ 'round\ up') \end{aligned}$$

(ii) $e = 0.02$

$$\begin{aligned} n &= \frac{z_{\frac{\alpha}{2}}^2}{4e^2} \\ &= \frac{z_{0.025}^2}{0.02^2} \\ &= \frac{1.96^2}{0.02^2} \\ &= 2401 \end{aligned}$$

EXAMPLE

In a survey of 300 company executives carried out by the Zambia Congress of Trade Union (ZCTU), 81 executives said they are willing to publicly disclose their annual salaries. Find a 99% confidence interval for the proportion of all executives who are willing to disclose their annual salaries.

SOLUTION

$n = 300,\ \ x = 81, \qquad 100(1-\alpha)\% = 99\% \implies 1-\alpha = 0.99 \ \therefore \alpha = 0.01$

$$\hat{p} \pm z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\frac{81}{300} \pm z_{0.005}\sqrt{\frac{\frac{81}{300}\left(1-\frac{81}{300}\right)}{300}}$$

$$0.27 \pm 2.575\sqrt{\frac{0.27(0.73)}{300}}$$

$$0.27 \pm 0.066002$$

$$0.203998 < p < 0.336002$$

Interpretation: Between 20.4 % and 33.6 % of all company executives are willing to publicly disclose their annual salaries.

C. ESTIMATING CONFIDENCE INTERVAL FOR A POPULATION VARIANCE BASED ON ONE SAMPLE

- ✓ Consider a random sample of size n drawn from a normal population with an unknown variance σ^2.
- ✓ A point estimator of σ^2 is S^2 i.e. s^2 is a point estimate of σ^2, where

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$= \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2\right]$$

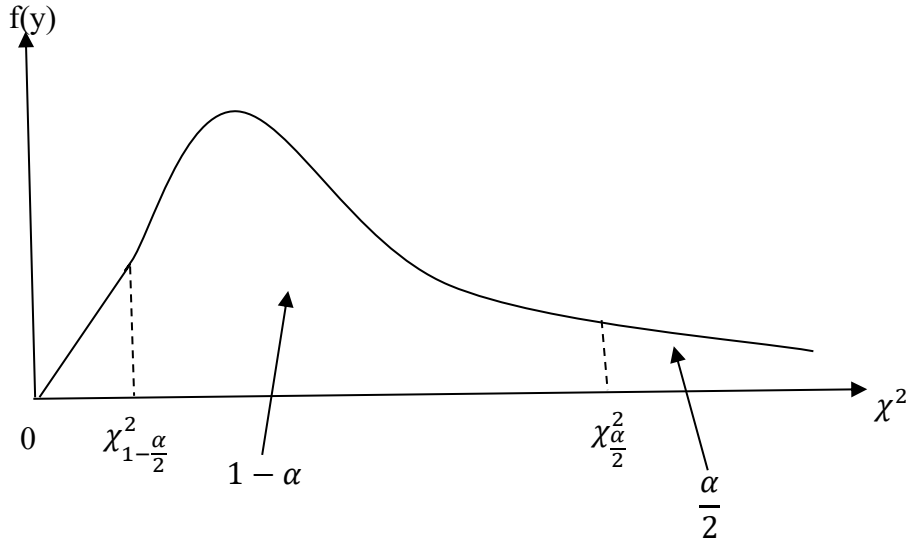
- ✓ We derive a confidence interval for the population variance σ^2 based on the chi- square distribution (χ^2 − distribution).
- ✓ In other words, a confidence interval for σ^2 is obtained from the statistic

$$Y = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

which follows a chi-square distribution with $n-1$ degrees of freedom.

NOTE

1. A chi-square distribution, unlike the normal and t − distributions, is not symmetric, is characterized by its degrees of freedom and it is always positive.
2. Reading a chi-square table is similar to reading a t − distribution table. Consider



i.e.

$$P\left(\chi^2_{1-\frac{\alpha}{2}} < Y < \chi^2_{\frac{\alpha}{2}}\right) = 1-\alpha$$

$$P\left(\chi^2_{1-\frac{\alpha}{2}} < \frac{(n-1)S^2}{\sigma^2} < \chi^2_{\frac{\alpha}{2}}\right) = 1-\alpha$$

$$P\left(\frac{1}{\chi^2_{1-\frac{\alpha}{2}}} > \frac{\sigma^2}{(n-1)S^2} > \frac{1}{\chi^2_{\frac{\alpha}{2}}}\right) = 1-\alpha$$

$$P\left(\frac{1}{\chi^2_{\frac{\alpha}{2}}} < \frac{\sigma^2}{(n-1)S^2} < \frac{1}{\chi^2_{1-\frac{\alpha}{2}}}\right) = 1-\alpha$$

$$P\left(\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}}}\right) = 1-\alpha$$

✓ Thus, if s^2 is the variance of a random sample of size n drawn from a normal population, a $100(1-\alpha)\%$ confidence interval for variance σ^2 is given by

$$\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2},n-1}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}$$

where $\chi^2_{\frac{\alpha}{2},n-1}$ and $\chi^2_{1-\frac{\alpha}{2},n-1}$ are χ^2 values with $n-1$ degrees of freedom leaving upper tail areas of $\frac{\alpha}{2}$ and $1-\frac{\alpha}{2}$ respectively.

- ✓ Or using the χ^2 − distribution with n − 1 degree of freedom we have the confidence interval for a population variance given by:

$$C.I: \left(\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2},n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2},n-1}} \right)$$

- ✓ **Interpretation**: we are $(1 - \alpha)100\%$ confident that the population variance σ^2 falls in the interval given by the equation above.

EXAMPLE

The following are weights, in grammes, of 10 packages of grass seed distributed by a certain company:

46.4 46.1 45.8 47.0 46.1 45.9 45.8 46.9 45.2 46.0

Assuming a normal population, find a 95% confidence interval for the variance of all such packages distributed by the company.

SOLUTION

$$n = 10\ , \alpha = 0.05$$

$$s^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n}\right]$$

$$= \frac{1}{9}\left[21273.12 - \frac{461.2^2}{10}\right]$$

$$= 0.286222$$

Note that this can again be obtained directly from your calculator, there is no need to show any working for this.

$$\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2},n-1}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}$$

$$\frac{9(0.286222)}{\chi^2_{0.025,9}} < \sigma^2 < \frac{9(0.286222)}{\chi^2_{0.975,9}}$$

$$\frac{9(0.286222)}{19.023} < \sigma^2 < \frac{9(0.286222)}{2.7}$$

$$0.135 < \sigma^2 < 0.954$$

NOTE

A $100(1-\alpha)\%$ confidence interval for σ is obtained by taking the square root of the endpoints of the confidence interval for σ^2 i.e.

$$\sqrt{0.135} < \sigma < \sqrt{0.954}$$

$$0.368 < \sigma < 0.977$$

2.4. TWO SAMPLE ESTIMATION

- ✓ Many problems in statistics involve comparing two parameters.

EXAMPLE

Suppose we want to study the effect of a course in reading comprehension. What do we do? Consider the following:

A. - Select n_1 students for the course, give them a test.
   - Select another group of n_2 students, give them a test.
   - Compare the scores for the two groups.

   OR

B. – select n students, give them a test before and after taking the course.
   – compare the scores

Which approach is better? Why? The second approach (B) is better because it reduces variability among the subjects.

In A we have 2 independent samples. In B we have 2 dependent samples.

DEFINITION

- ✓ Two samples are independent if the data values obtained from one are unrelated to the values from the other. The samples are dependent if each data value from one sample is paired in a natural way with a data value from the other sample.

A. TWO POPULATION MEANS (DEPENDENT SAMPLES)

✓ Suppose we want to estimate the difference between two population means

$\mu_D = \mu_1 - \mu_2$ where μ_1 and μ_2 are population means of X_1 and X_2 respectively.

We estimate μ_D by $\bar{d}$ where

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i \ \ , d_i = x_{1i} - x_{2i}$$

i.e. d_i are differences between paired observations.

✓ Thus, if $\bar{d}$ and s_d are the mean and standard deviation of normally distributed differences of n random pairs of measurements, a $100(1-\alpha)\%$ confidence interval for $\mu_D = \mu_1 - \mu_2$ is

$$\bar{d} \pm t_{\frac{\alpha}{2},n-1}\frac{s_d}{\sqrt{n}}$$

where $t_{\frac{\alpha}{2},n-1}$ is t – value leaving an upper tail area of $\frac{\alpha}{2}$.

EXAMPLE

Agricultural departments of 9 universities took part in a study to test the yield capabilities of two new varieties of wheat. Each variety was planted on plots of equal area at each university and the yields, in kilogrammes per plot, recorded as follows:

	University								
Variety	1	2	3	4	5	6	7	8	9
1	38	23	35	41	44	29	37	31	38
2	45	25	31	38	50	33	36	40	43

Assume that the differences in the yields are normally distributed.

(a) Find a 95% confidence interval for the mean difference between the yields of the varieties.

(b) Is there a difference in the mean yield of the two varieties?

SOLUTION

(a) $d_i = x_1 - x_2$ $(i.e. variety\ 1 - variety\ 2\ at\ each\ university)$

-7, -2, 4, 3, -6, -4, 1, -9, -5

$\bar{d} = -2.78, \ \ s_d = 4.5765, n = 9, \alpha = 0.05$

($\bar{d}$ and s_d are obtained directly from the calculator)

$$\bar{d} \pm t_{\frac{\alpha}{2},n-1}\frac{s_d}{\sqrt{n}}$$

$$-2.78 \pm t_{0.025,8}\frac{4.5765}{\sqrt{9}}$$

$$-2.78 \pm 2.306\left(\frac{4.5765}{3}\right)$$

$$(-6.3, 0.7)\ or\ -6.3 < \mu_1 - \mu_2 < 0.7$$

Note that if $d_i = x_2 - x_1$ then the confidence interval would be $(-0.7, 6.3)$

(b) The confidence interval contains 0, therefore there's no difference in the mean yield of the two varieties.

Note that if there is no difference between the varieties then the mean difference would be 0, that is why we conclude that there is no difference if 0 is in the confidence interval.

B. TWO POPULATION MEANS (INDEPENDENT SAMPLES)

- ✓ If we have two independent samples of sizes n_1 and n_2 from populations X_1 and X_2, then an estimate for $\mu_1 - \mu_2$ is $\bar{x}_1 - \bar{x}_2$ ($\bar{X}_1 - \bar{X}_2$ is the estimator).
- ✓ To construct a $100(1-\alpha)\%$ confidence interval of two sample means we consider three cases:

Case I: When the population variances are known i.e., $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ where σ_1^2 and σ_2^2 are known then

$$C.I: (\bar{x}_1 - \bar{x}_2) \pm z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

NOTE

1. It doesn't matter whether the samples n_1 and n_2 are large or small the interval is based on the normal distribution.
2. This is an exact confidence interval.

Case II: When the population variances are not known but equal (i.e., $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$ but σ_1^2 and σ_2^2 are unknown)

(a) Assume that $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (i.e. the unkown variances are equal). Estimate σ^2 by the pooled variance given by:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

then

$$C.I: (\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2},\nu}\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$$

where $\nu = n_1 + n_2 - 2$ is the degrees of freedom for the t – distribution.

(b) When the population variances are not known and not equal (i.e., if $\sigma_1^2 \neq \sigma_2^2$) then

$$C.I: (\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2},\nu}\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}$$

where $t_{\frac{\alpha}{2},\nu}$ is a t – distribution with

$$\nu = \frac{\left(\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1}+\frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$$

degrees of freedom.

NOTE

1. This case only applies to small samples.
2. Part(a) of this case gives an exact confidence interval while part(b) gives an approximate confidence interval.

Case III: The distributions of X_1 and X_2 are unknown but the samples are large ($n_1 \geq 30,\ n_2 \geq 30$)

(a) If σ_1^2 and σ_2^2 are known then

$$C.I: (\bar{x}_1 - \bar{x}_2) \pm z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}$$

(b) If σ_1^2 and σ_2^2 are unknown then

$$C.I: (\bar{x}_1 - \bar{x}_2) \pm z_{\frac{\alpha}{2}}\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}$$

NOTE

1. For this case the confidence intervals are approximate.
2. If the sample sizes are small the populations have to be normal and Cases (I) and (II) will apply. The normal assumption is not required if the samples are large and Case (III) will apply.

Interpretation: we are $100\ (1\ -\ \alpha)$ % confident that the difference in population means $(\mu_1\ -\ \mu_2)$ falls in the interval given by the equations for large samples as well as for small samples.

EXAMPLES

1. A study was conducted to compare gas mileage of two types of engines A and B. Fifty experiments conducted using type A engine gave an average gas mileage of 36 miles per gallon while 75 experiments conducted using type B engine resulted in an average gas mileage of 42 miles per gallon. Assume that the population standard deviations are 6 and 8 for machines A and B, respectively.
   (a) Find a 96% confidence interval for $\mu_B - \mu_A$, where μ_B and μ_A are the population mean gas mileage for machines B and A respectively.
   (b) Are the two types of engines different in terms of gas mileage?

   SOLUTION

   (a) $n_A = 50\ \ \bar{x}_A = 36\ \ \sigma_A = 6$

   $n_B = 75\ \ \bar{x}_B = 42\ \ \sigma_B = 8$

   (Note that the sample sizes are large, so we use Case III), $\alpha = 0.04$

$$\bar{x}_B - \bar{x}_A \pm z_{\frac{\alpha}{2}}\sqrt{\frac{\sigma_B^2}{n_B} + \frac{\sigma_A^2}{n_A}}$$

$$42 - 36 \pm z_{0.02}\sqrt{\frac{8^2}{75} + \frac{6^2}{50}}$$

$$6 \pm 2.05\sqrt{\frac{64}{75} + \frac{36}{50}}$$

$$(3.43, 8.57)\ or\ 3.43 < \mu_B - \mu_A < 8.57$$

   Note that $P(Z < 2.05) = 0.9798$ and $P(Z < 2.06) = 0.9803$, 0.98 is between 0.9798 and 0.9803 but closer to 0.9798 so we go for 2.05

   (b) The engines are different in terms of gas mileage since the confidence interval doesn't contain 0.

2. In an air pollution study, a town engineer took a sample of carbon monoxide measurements near a power plant to compare with another sample obtained from other locations in the town. The results were as follows:

Near plant: 40 16 44 47 36 64 35 53 45 52 31 38 44 29 45 47

Other locations: 34 28 15 43 26 27 35 9 32 25 27 35 44 18

Find a 90% confidence interval for the difference between the population means, assuming that the populations are normally distributed with equal variances.

SOLUTION

(a) Let $\mu_1 = mean\ near\ plant$ and $\mu_2 = mean\ at\ other\ locations$

$$n_1 = 16 \quad \bar{x}_1 = 41.625 \quad s_1 = 11.104804$$

$$n_2 = 14 \quad \bar{x}_2 = 28.429 \quad s_2 = 9.897275$$

(Note that the sample sizes are small, population standard deviations are unkown but can be assumed equal so it is Case II (a). Note also that the sample means and standard deviations are directly obtained from the calculator).

$$\alpha = 0.1$$

$$C.I: \bar{x}_1 - \bar{x}_2 \pm t_{\frac{\alpha}{2},n_1+n_2-2}\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$$

$$\begin{aligned} s_p^2 &= \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} \\ &= \frac{15(11.104804)^2 + 13(9.897275)^2}{16+14-2} \\ &= 111.54210 \end{aligned}$$

$$41.625 - 28.429 \pm t_{0.05,28}\sqrt{111.54210\left(\frac{1}{16}+\frac{1}{14}\right)}$$

$$13.196 \pm 1.701\sqrt{111.54210\left(\frac{1}{16}+\frac{1}{14}\right)}$$

$$6.62 < \mu_1 - \mu_2 < 19.77$$

3. The amount of phosphorus was measured (in mg/litre) at two stations along the Kafue river. Fifteen samples from station 1 had an average phosphorus content of 3.84 mg/litre and a standard deviation of 3.07 mg/litre, while 12 samples from station 2 had an average content of 1.49 mg/litre and a standard deviation of 0.80 mg/litre. Find a 95% confidence interval for the difference in the true phosphorus

contents at these two stations, assuming that the populations came from normal populations.

SOLUTION

$$n_1 = 15 \quad \bar{x}_1 = 3.84 \quad s_1 = 3.07$$
$$n_2 = 12 \quad \bar{x}_2 = 1.49 \quad s_2 = 0.80$$
$$\alpha = 0.05$$

(Note that the sample sizes are small, population standard deviations unknown and nothing is mentioned about assuming equality of variances, so it is Case II(b))

$$C.I: \bar{x}_1 - \bar{x}_2 \pm t_{\frac{\alpha}{2},\nu}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$
$$= \frac{\left(\frac{3.07^2}{15} + \frac{0.8^2}{12}\right)^2}{\frac{\left(\frac{3.07^2}{15}\right)^2}{14} + \frac{\left(\frac{0.8^2}{12}\right)^2}{12}}$$
$$= 16.3$$
$$\approx 16$$

$$3.84 - 1.49 \pm t_{0.025,16}\sqrt{\frac{3.07^2}{15} + \frac{0.8^2}{12}}$$

$$2.35 \pm 2.12\sqrt{\frac{3.07^2}{15} + \frac{0.8^2}{12}}$$

$$(0.60, 4.10)$$

C. TWO PROPORTIONS

- ✓ When population proportions p_1 and p_2 are known then a $100(1 - \alpha)$ % $C.I$ is estimated around the differences between the two sample proportions.
- ✓ If we have two independent samples of sizes n_1 and n_2 from populations X_1 and X_2, then an estimate for $p_1 - p_2$ is $\hat{p}_1 - \hat{p}_2$ ($\hat{P}_1 - \hat{P}_2$ is the estimator) where

$\hat{p}_1 = \frac{x_1}{n_1}$, $x_1 =$ number of subjects with characteristic of interest in sample 1

$\hat{p}_2 = \frac{x_2}{n_2}$, $x_2 =$ number of subjects with characteristic of interest in sample 2

NOTE

It can be shown that

1. $E(\hat{P}_1 - \hat{P}_2) = \mu_{\hat{P}_1 - \hat{P}_2} = p_1 - p_2 (i.e$ from $MAT2901$ if $X \sim Bin(n,p)$, then $E(X) = np)$

$\Rightarrow E(\hat{P}_1 - \hat{P}_2) = E\left(\frac{X_1}{n_1} - \frac{X_2}{n_2}\right) = \frac{E(X_1)}{n_1} - \frac{E(X_2)}{n_2} = \frac{n_1 p_1}{n_1} - \frac{n_2 p_2}{n_2} = p_1 - p_2$ and

2. $Var(\hat{P}_1 - \hat{P}_2) = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$ from $MAT2901$ if $X \sim Bin(n,p)$, then $Var(X) = np(1-p)$.

$\Rightarrow Var(\hat{P}_1 - \hat{P}_2) = Var\left(\frac{X_1}{n_1} - \frac{X_2}{n_2}\right) = \frac{Var(X_1)}{n_1^2} + \frac{Var(X_2)}{n_2^2}$

$= \frac{n_1 p_1(1-p_1)}{n_1^2} + \frac{n_2 p_2(1-p_2)}{n_2^2} = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$

3. $\hat{P}_1 - \hat{P}_2 \sim N\left(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right)$ $(approximately)$

✓ To construct a $100(1-\alpha)\%$ for two sample proportions, we consider two cases:

Case I: when two samples are large $(i.e., n_1, n_2 \geq 30)$

Thus, for large samples if $\hat{p}_1$ and $\hat{p}_2$ are the proportion of successes in random samples of sizes n_1 and n_2 from independent populations X_1 and X_2, respectively, then an approximate $100(1-\alpha)\%$ confidence interval estimate for $p_1 - p_2$ is given by

$$C.I: (\hat{p}_1 - \hat{p}_2) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

NOTE

When population proportions p_1, p_2 are unknown, then the $C.I$ is give by either

1. $C.I: (p_1 - p_2) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$ or

2. $C.I: (p_1 - p_2) \pm z_{\frac{\alpha}{2}} \sqrt{\frac{1}{4n_1} + \frac{1}{4n_2}}$

Interpretation: we are 100 $(1 - \alpha)$ % confident that the in the population proportions $(p_1 - p_2)$ falls in the interval given by equation three equations.

Case II: When sample sizes are small $(i.e., n_1 + n_2 < 30)$.

When samples are small, we use the t −distribution.

Thus, for small samples the confidence interval estimates for $(p_1 - p_2)$ is given by:

$$C.I: (\hat{p}_1 - \hat{p}_2) \pm t_{\frac{\alpha}{2}} (n_1 + n_2 - 2) \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

EXAMPLE

A mobile phone dealer conducted a study to ascertain market accessibility of a new cellular phone they wished to market. The study was conducted in the two major cities of Zambia Lusaka and Ndola. In Lusaka 300 shoppers were interviewed and 210 were impressed by the new phone while in Ndola of 250 shoppers who were interviewed, 80 were impressed by the new phone. Construct a 95 % confidence interval of the difference between the actual proportion of Lusaka and Ndola shoppers who were impressed by the new cellular phone.

SOLUTION

Sample 1 (Lusaka)	Sample 2 (Ndola)
$n_1 = 300$	$n_2 = 250$
$\hat{p}_1 = \frac{210}{300} = 0.7$	$\hat{p}_2 = \frac{80}{250} = 0.32$
$1 - \hat{p}_1 = 0.3$	$1 - \hat{p}_2 = 0.68$

Since the combined sample size is large, a 95% confidence interval for $(p_1 - p_2)$ is given by:

$$C.I: (0.7 - 0.32) \pm z_{\frac{0.05}{2}} \sqrt{\frac{0.7(0.3)}{300} + \frac{0.32(0.68)}{250}}$$

$$0.38 \pm z_{0.025}\sqrt{\frac{0.21}{300}+\frac{0.2176}{250}}$$

$$0.38 \pm 1.96(0.0396)$$

$$0.38 \pm 0.0777$$

$$(0.3023, 0.4577) \text{ or } 0.3023 < p_1 - p_2 < 0.4577$$

Interpretation: The 95 % confidence interval ranges from 30% to 46%.

The confidence interval does not contain zero implying that there is a 95% chance that the difference on the true proportion of shoppers in Lusaka and Ndola is quite significant.

EXAMPLE

Refer to the previous example above, suppose 15 Lusaka shoppers and 10 Ndola shoppers were interviewed and that 9 Lusaka shopper liked the cellular phone compared to 4 Ndola shoppers. Construct a 95 % confidence interval of the difference between the actual proportion of Lusaka and Ndola shoppers who were impressed by the new cellular phone**.**

SOLUTION

Sample 1 (Lusaka)	Sample 2 (Ndola)
$n_1 = 15$	$n_2 = 10$
$\hat{p}_1 = \frac{9}{15} = 0.6$	$\hat{p}_2 = \frac{4}{10} = 0.4$
$1 - \hat{p}_1 = 0.4$	$1 - \hat{p}_2 = 0.6$

Since samples are small $(n_1 + n_2 < 30)$ we use a t-distribution. A 95% confidence interval for $(p_1 - p_2)$ is given by:

$$C.I\text{: } (\hat{p}_1 - \hat{p}_2) \pm t_{\frac{\alpha}{2}}(n_1 + n_2 - 2)\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1}+\frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$0.6 - 0.4 \pm t_{0.025}\ (24)\sqrt{\frac{0.6(0.4)}{15}+\frac{0.4(0.6)}{10}}$$

$$0.2 \pm 2.06(0.2)$$

$$0.2 \pm 0.412$$

$$(-0.212, 0.612) \text{ or } -0.212 < p_1 - p_2 < 0.612$$

Interpretation: There is a 95% probability that the true difference in the proportion of Lusaka and Ndola shoppers who liked the cellular phone lies between −0.212 and 0.612. The confidence interval covers zero, therefore, there is a 95% chance that there is no difference in the true proportion of shoppers in Lusaka and Ndola who liked the new phone

EXAMPLE

A cigarette – manufacturing firm claims that its brand A line of cigarettes outsells its brand B line by 8%. If it is found that 42 of 200 smokers prefer brand A and 18 of 150 smokers prefer brand B. Compute a 94% confidence interval for the difference between the proportions of sales of the two brands. Is the claim valid?

SOLUTION

$$n_A = 200 \quad x_A = 42 \quad \hat{p}_A = \frac{x_A}{n_A} = \frac{42}{200} = 0.21$$

$$n_B = 150 \quad x_B = 18 \quad \hat{p}_B = \frac{x_B}{n_B} = \frac{18}{150} = 0.12$$

$$\alpha = 0.06$$

$$\hat{p}_A - \hat{p}_B \pm z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{n_A} + \frac{\hat{p}_B(1-\hat{p}_B)}{n_B}}$$

$$0.21 - 0.12 \pm z_{0.03} \sqrt{\frac{0.21(1-0.21)}{200} + \frac{0.12(1-0.12)}{150}}$$

$$0.09 \pm 1.88 \sqrt{\frac{0.21(1-0.21)}{200} + \frac{0.12(1-0.12)}{150}}$$

$$(0.016, 0.164) \; or \; 0.016 < p_A - p_B < 0.164$$

0.08 lies in the interval, therefore the claim is valid.

D. ESTIMATING A RATIO OF TWO VARIANCES

- ✓ A point estimate of the ratio of two population variances $\frac{\sigma_1^2}{\sigma_2^2}$ is $\frac{s_1^2}{s_2^2}$ (i.e. $\frac{s_1^2}{s_2^2}$ is the estimator). If σ_1^2 and σ_2^2 are variances of normal populations, then a confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$ is obtained from the statistic

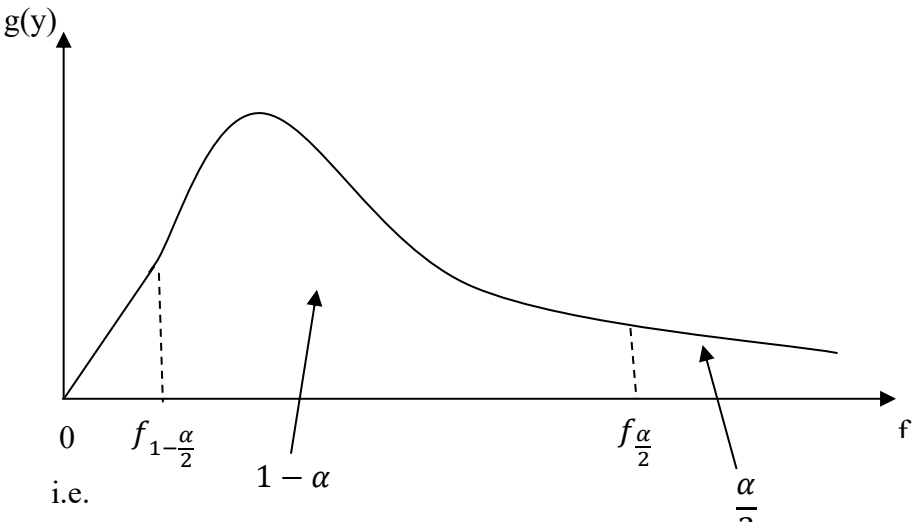
$$F = \frac{\frac{s_1^2}{\sigma_1^2}}{\frac{s_2^2}{\sigma_2^2}} = \frac{\sigma_2^2 s_1^2}{\sigma_1^2 s_2^2} \sim F(\nu_1, \nu_2)$$

which follows an F – distribution with $\nu_1 = n_1 - 1$ numerator and $\nu_2 = n_2 - 1$ denominator degrees of freedom.

NOTE:

1. The F – distribution is defined by numerator and denominator degrees of freedom.
2. The F – distribution is always positive and shape of the distribution depends on the degrees of freedom.

Consider



i.e.

$$P\left(f_{1-\frac{\alpha}{2}}(\nu_1, \nu_2) < F < f_{\frac{\alpha}{2}}(\nu_1, \nu_2)\right) = 1 - \alpha$$

$$P\left(f_{1-\frac{\alpha}{2}}(\nu_1, \nu_2) < \frac{\sigma_2^2 s_1^2}{\sigma_1^2 s_2^2} < f_{\frac{\alpha}{2}}(\nu_1, \nu_2)\right) = 1 - \alpha$$

$$P\left(\frac{1}{f_{1-\frac{\alpha}{2}}(\nu_1, \nu_2)} > \frac{\sigma_1^2 s_2^2}{\sigma_2^2 s_1^2} > \frac{1}{f_{\frac{\alpha}{2}}(\nu_1, \nu_2)}\right) = 1 - \alpha$$

$$P\left(\frac{1}{f_{\frac{\alpha}{2}}(\nu_1, \nu_2)} < \frac{\sigma_1^2 s_2^2}{\sigma_2^2 s_1^2} < \frac{1}{f_{1-\frac{\alpha}{2}}(\nu_1, \nu_2)}\right) = 1 - \alpha$$

$$P\left(\frac{s_1^2}{s_2^2}\frac{1}{f_{\frac{\alpha}{2}}(\nu_1, \nu_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2}\frac{1}{f_{1-\frac{\alpha}{2}}(\nu_1, \nu_2)}\right) = 1 - \alpha$$

✓ Thus, if s_1^2 and s_2^2 are variances of independent samples of sizes n_1 and n_2, respectively, from normal populations, then a $100(1-\alpha)\%$ confidence interval for $\frac{\sigma_1^2}{\sigma_2^2}$ is given by

$$\frac{s_1^2}{s_2^2}\frac{1}{f_{\frac{\alpha}{2}}(\nu_1,\nu_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2}\frac{1}{f_{1-\frac{\alpha}{2}}(\nu_1,\nu_2)}$$

where $f_{\frac{\alpha}{2}}(\nu_1,\nu_2)$ is an f – value with $\nu_1 = n_1 - 1$ numerator and $\nu_2 = n_2 - 1$ denominator degrees of freedom leaving an upper tail area of $\frac{\alpha}{2}$.

NOTE

The F – distribution table only gives f – values for small upper tail probabilities such as $\alpha = 0.1,\ \alpha = 0.05\ etc$. To find f – values for big upper tail areas such as $\alpha = 0.95$ we use the following property

$$f_{\alpha}(\nu_1,\nu_2) = \frac{1}{f_{1-\alpha}(\nu_2,\nu_1)}$$

EXAMPLE

A college mathematics department was considering two screening exams for the purpose of placing students. Students in two sections of an introductory maths course were given the test and the following results were obtained:

Exam	n	s^2
A	21	121
B	16	64

(a) Find a 95% confidence interval for

(i) $\frac{\sigma_A^2}{\sigma_B^2}$

(ii) $\frac{\sigma_A}{\sigma_B}$

(b) Find a 90% confidence interval for $\frac{\sigma_B^2}{\sigma_A^2}$.

(c) Are the population variances of the two exams different?

SOLUTION

(a) Let $\sigma_1^2 = \sigma_A^2$, $\sigma_2^2 = \sigma_B^2$ then

(i) $n_1 = 21 \quad s_1^2 = 121$

$n_2 = 16 \quad s_2^2 = 64, \alpha = 0.05$

$$\frac{s_1^2}{s_2^2}\frac{1}{f_{\frac{\alpha}{2}}(n_1-1,n_2-1)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2}\frac{1}{f_{1-\frac{\alpha}{2}}(n_1-1,n_2-1)}$$

$$\frac{121}{64}\frac{1}{f_{0.025}(20,15)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{121}{64}\frac{1}{f_{0.975}(20,15)}$$

$$\frac{121}{64}\frac{1}{2.76} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{121}{64}f_{0.025}(15,20)$$

$$0.68501 < \frac{\sigma_1^2}{\sigma_2^2} < \frac{121}{64}(2.57)$$

$$0.68501 < \frac{\sigma_1^2}{\sigma_2^2} < 4.85891$$

$$0.68501 < \frac{\sigma_A^2}{\sigma_B^2} < 4.85891$$

(ii)

$$\sqrt{0.68501} < \frac{\sigma_A}{\sigma_B} < \sqrt{4.85891}$$

$$0.828 < \frac{\sigma_A}{\sigma_B} < 2.204$$

(b) $\alpha = 0.1$

$$\frac{s_2^2}{s_1^2}\frac{1}{f_{\frac{\alpha}{2}}(n_2-1,n_1-1)} < \frac{\sigma_2^2}{\sigma_1^2} < \frac{s_2^2}{s_1^2}\frac{1}{f_{1-\frac{\alpha}{2}}(n_2-1,n_1-1)}$$

$$\frac{64}{121}\frac{1}{f_{0.05}(15,20)} < \frac{\sigma_2^2}{\sigma_1^2} < \frac{64}{121}\frac{1}{f_{0.95}(15,20)}$$

$$\frac{64}{121}\frac{1}{2.2} < \frac{\sigma_2^2}{\sigma_1^2} < \frac{64}{121}f_{0.05}(20,15)$$

$$0.240 < \frac{\sigma_2^2}{\sigma_1^2} < \frac{64}{121}(2.33)$$

$$0.240 < \frac{\sigma_2^2}{\sigma_1^2} < 1.232$$

$$0.240 < \frac{\sigma_B^2}{\sigma_A^2} < 1.232$$

(c) The population variances of the two exams are not different since the confidence interval contains 1. (Note that the ratio of the variances would be 1 if the variances were equal.)

3. STATISTICAL HYPOTHESIS TESTING

3.1. INTRODUCTION

- ✓ Hypothesis testing is that branch of statistical inference that is used to verify claims made concerning population parameters.
- ✓ Hypothesis testing is important because it provides an objective framework for making decisions using probabilistic methods rather than relying on subjective impressions.
- ✓ The aim of hypothesis testing is to test if certain claims can be made about a population based on the data from a sample.
- ✓ ***Definition*** **(**Statistical hypothesis**)**: A statistical hypothesis is an assertion or conjecture concerning one or more populations.
- ✓ In other words, a statistical hypothesis is a claim or a guess or an assumption or a statement, which may or may not be true, made concerning a population parameter.
- ✓ Note that a hypothesis is usually stated in terms of a population parameter.

3.2. TYPES OF HYPOTHESIS

- ✓ There are two types of hypotheses namely, the null hypothesis (denoted by H_0) and the alternative hypothesis (denoted by H_1 or H_a).
- ✓ ***Definition*** (null hypothesis): The null hypothesis is a statement asserting no change or no difference or no effect.
- ✓ It usually takes the form of a statement about a population parameter (or parameters) containing an equal sign.
- ✓ In general, a **null hypothesis** is an assertion about the value of a population parameter or it is a formal statement of a claim being made concerning the population parameter
- ✓ ***Definition*** (alternative hypothesis): The alternative hypothesis is a statement that might be true instead of the null.
- ✓ In other words, it is the negation of the null hypothesis. It usually contains the sign $>$ or $<$ or $\neq$.
- ✓ Note that a choice is made between the null hypothesis (H_0) and the alternative hypothesis (H_1) based on information given by a sample,
- ✓ For example, the null hypothesis might assert that the population mean is equal to a specified value μ_0.

- ✓ This can be written as: $\boldsymbol{H_0: \mu = \mu_0}$**.** However, the alternative hypothesis opposes this assertion and it is written as: $\boldsymbol{H_1: \mu \neq \mu_0}$.
- ✓ In this case, the alternative hypothesis suggests that the mean takes values that are either below μ_0 or above it.
- ✓ Thus, to investigate H_0 we conduct a non-directional test which is known as a **two-tailed test**.
- ✓ The null hypothesis might assert that the population mean is at least equal to a certain specified value μ_0 that is, $\boldsymbol{H_0: \mu \geq \mu_0}$.
- ✓ In this case, the alternative hypothesis would consist of values below μ_0 that is $\boldsymbol{H_1: \mu < \mu_0}$.---------- (i)
- ✓ Similarly, if the null hypothesis asserts that the population mean is less than or equal to a certain specified value μ_0 that is, $\boldsymbol{H_0: \mu \leq \mu_0}$, then the alternative will be

  $\boldsymbol{H_1: \mu > \mu_0}$.---------- (ii)
- ✓ Note that since in both cases of (i) and (ii), the alternative hypotheses consist of values either below or above the specified value μ_0, we conduct a **one-sided test** or a **one-tailed test** to investigate H_0.

  NOTE

  (i) A **one-sided test** or **one-tailed test** is a test of a statistical hypothesis where the region of rejection is on only one side of the sampling distribution.

  (ii) A **two-tailed test** or a **two-sided test** is a test of a statistical hypothesis where the region of rejection is on both sides of the sampling distribution.

3.3. HYPOTHESIS TESTING

- ✓ Hypothesis testing involves gathering evidence from a random sample drawn from a population of interest in order to decide whether the null hypothesis is likely to be true or false.
- ✓ The hypothesis is rejected if evidence from the sample is not consistent with the stated hypothesis, otherwise it is accepted.
- ✓ However, the acceptance of the stated hypothesis does not necessarily imply that it is true, rather it is the result of insufficient evidence to reject it.

  NOTE

  1. The procedure for choosing between hypotheses is called hypothesis testing.

2. In hypothesis testing, the idea is to give the benefit of the doubt to the null hypothesis.
3. The null hypothesis will be rejected (and the alternative accepted) only if the sample data suggest beyond any reasonable doubt that the null hypothesis is false.

3.4. DECIDING ON THE NULL HYPOTHESIS

- ✓ Determining what a null hypothesis should be in a given situation may prove to be difficult.
- ✓ However, if the null hypothesis is wrongly formulated, then the test will be meaningless.
- ✓ The following steps will be convenient in deciding what a null hypothesis should be:

1. The null hypothesis always has an element of equality, that is, either an equal to sign $(=)$ or a greater than or equal to sign $(\geq)$ or a less than or equal to sign $(\leq)$ is used in the expression of H_0.
2. The null hypothesis is usually an expression of a claim made by someone. However, if a claim does not include an equal to sign $(=)$ it becomes the alternative hypothesis.
3. The null hypothesis is the hypothesis we formulate with the hope of rejecting.
4. If the null hypothesis is true, then no corrective action would be necessary, whereas if the alternative hypothesis is true some corrective action would be necessary.

- ✓ Note that when choosing H_0 and H_1, usually the hypothesis will take the form of a claim, a belief or a suspicion.

EXAMPLES

1. A nutritionist claimed that a food company's cans of soup contained more than 900mg of sodium on average. We test the hypotheses

SOLUTION

$$H_0: \mu = 900$$
$$H_1: \mu > 900 \quad \leftarrow claim$$

2. An auto company claims that the mean weight of its pickup trucks is 2 tonnes. Test the hypotheses

SOLUTION

$$H_0: \mu = 2 \leftarrow claim$$
$$H_1: \mu \neq 2$$

3. The proportion of female students at UNZA is less than 0.4 i.e.

SOLUTION

$$H_0: p = 0.4 \leftarrow claim$$
$$H_1: p < 0.4$$

4. A company director claims that the average age of an employee at his company is 21. A new programme coordinator at the company doubts this claim. Set up the null and alternative hypotheses if the coordinator wishes to show that it is not 21.

SOLUTION

$$H_0: \mu = 21 \leftarrow claim$$
$$H_1: \mu \neq 21 \ (two - \text{tailed test})$$

5. A leading bakery in Zambia claims that the average cost of producing a standard loaf of bread is k100.00. If you suspect that the claims exaggerate the costs, how would you set up the null and alternative hypothesis?

SOLUTION

$$H_0: \mu \geq \text{k}100$$
$$H_1: \mu < \text{k}100 \ (\text{one} - \text{tailed test})$$

6. A food company produces packets of peanuts weighing 336g (on the average). Periodically, the quality control department takes samples of peanut packets to determine whether the packaging process is under control.

SOLUTION

The hypotheses of interest are

$$H_0: \mu = 336g \ (process\ is\ under\ control)$$
$$H_1: \mu \neq 336g \ (process\ is\ not\ under\ control)$$

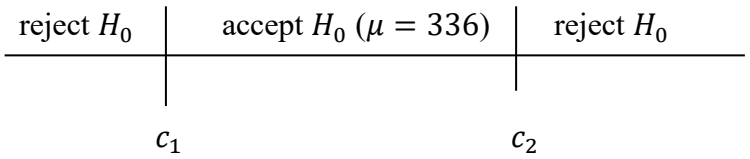
To decide between the two hypotheses, we

1. Choose a random sample of size n.
2. Calculate $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$.
3. Reject H_0 if $\bar{x}$ is "too large" or "too small" compared to 336.
4. Find how large or small should $\bar{x}$ be, by finding c_1 and c_2 such that

if $\bar{x} < c_1$ reject H_0

if $c_1 \leq \bar{x} \leq c_2$ accept H_0

if $\bar{x} > c_2$ reject H_0



c_1 and c_2 are called critical points. $(-\infty, c_1)$ and (c_2, ∞) constitute what is called a critical region, $[c_1, c_2]$ is the acceptance region.

Note:

1. Examples 1 and 3 are known as one sided test (or alternatives) while 2 is an example of a two-sided test.
2. The direction of the alternatives in Examples 1 and 3 are clearly given in the statement. The sign used in the alternative for Example 2 is simply the opposite of the null hypothesis i.e. if we reject the claim that the $\mu = 2$ then it should be $\mu \neq 2$.

3.5. TYPES OF ERRORS MADE IN HYPOTHESIS TESTING

- ✓ In deciding to reject or accept a null hypothesis, there will be chances of incorrectly rejecting or accepting it.
- ✓ In other words, rejecting and accepting the null hypothesis will result to errors in hypothesis testing and such errors may be due to faulty in sampling procedures.
- ✓ Thus, there are two types of errors that can be made in hypothesis testing, these include; type I and type II errors.
- ✓ **DEFINITION** (Type I error): A type I error is the rejection of the null hypothesis (H_0) when it is true.
- ✓ In other words, a type I error is committed when the true null hypothesis is rejected.
- ✓ **DEFINITION** (Type II error): A type II error is the acceptance of the null hypothesis (H_0) when it is false (not true).
- ✓ In other words, a type II error is committed if we accept the null hypothesis when it is false.

NOTE

1. The probability of committing a type I error is called the level of significance, and it is denoted by α. Thus, we represent the probability of a type I error by α, that is, $\alpha = P(Type\ I\ error) = P(rejecting\ H_0 \mid H_0\ is\ true)$
2. The probability of committing a type II error is denoted by β. Thus, we represent the probability of a type II error by β, that is
   $\beta = P(Type\ II\ error) = P(accepting\ H_0 \mid H_0\ is\ false)$

3. We can summarise the possible situations as follows:

		Reality	
		H_0 is true	H_0 is false
Decision	Reject H_0	Type I error (α)	Correct decision
	Accept H_0	Correct decision	Type I error (β)

4. $1-\beta$ is the probability of rejecting H_0 when H_0 is false. Thus, we the probability $1-\beta$ to as the power of the test or decision rule.
5. In the above example, a type I error is made if it is decided that the process in not under control ($i.e.\ \mu \neq 336g$) when infact it is under control ($i.e.\ \mu = 336g$). A type II error is committed if it is decided the process is under control when it is not.

3.6. STEPS IN HYPOTHESIS TESTING

✓ The following steps should be followed when conducting a hypothesis testing:

1. **Testing problem** (State the null hypothesis and the alternative hypothesis)

Here we formulate the null and alternative hypothesis, that is, we state the null hypothesis H_0 and the alternative hypothesis H_1 or H_a.

2. Test statistic

Here we identify the distribution and a test statistic to be used in the hypothesis

NOTE:

1. We use the $Z-distribution$ when testing for the population mean μ if:
   (i) The population standard deviation σ is known regardless of the sample size.
   (ii) The population standard deviation is unknown and n is large
2. We use the $t-distribution$ when the population standard deviation σ is unknown and the sample size n is small.

3. Critical region

Here we determine the critical region (i.e. the rejection and acceptance regions) and clearly indicate whether it is one- or two-sided test. Again, depending on the distribution identified and the level of significant desired, we find the value from statistical tables which we call a critical value.

NOTE:

1. The critical value is the value that separates the acceptance region from the rejection region.
2. The region of acceptance is a range of values such that if the test statistic falls within it, the null hypothesis is not rejected. In other words, the region of acceptance is defined so that the chance of making a type I error is equal to the significance level.
3. The rejection region is made up of the range of values such that if the test statistic calculated from sample data falls in it, the null hypothesis would be rejected.in other words, the rejection region is the set of values outside the region of acceptance

4. Observed value

Here we calculate the observed value of the test statistic and determine if it falls in the critical region.

NOTE:

1. A test statistic is a value calculated from sample data that is used to decide whether or not to reject.
2. Once the test statistic falls within the rejection region, is rejected

5. Conclusion

Here we make a decision based on the null hypothesis and later state the conclusion of the hypothesis testing.

NOTE:

1. If is rejected, we conclude that is true or we accept the alternative.
2. If we fail to reject, we conclude that the evidence gathered is insufficient to warrant the rejection of the null hypothesis.

3.7. TESTS ABOUT POPULATION MEANS

✓ Tests concerning population means are based on the sampling distribution of $\bar{x}$ or $\bar{x}_1 - \bar{x}_2$ assuming the null hypothesis H_0 is true.

(A) ONE POPULATION MEAN (OR THE POPULATION MEAN OF A SINGLE POPULATION)

✓ To test the hypothesis for a single population-mean such as the following:

$$H_0: \mu = \mu_0 \qquad H_0: \mu = \mu_0 \qquad H_0: \mu = \mu_0$$
$$H_1: \mu > \mu_0 \ \ or \ \ H_a: \mu < \mu_0 \ \ or \ \ H_A: \mu \neq \mu_0$$

we use the following test statistics:

1. If $X \sim N(\mu, \sigma^2)$ use

   (a) $Z = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$ if σ^2 is known

   (b) $T = \frac{\bar{x}-\mu_0}{\frac{s}{\sqrt{n}}} \sim T_{n-1}$ if σ^2 is unknown (for $n < 30$)

2. If X has a mean μ and variance σ^2 and $n \geq 30$, use

   (a) $Z = \frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}}$ if σ^2 is known

   (b) $Z = \frac{\bar{x}-\mu_0}{\frac{s}{\sqrt{n}}}$ if σ^2 is unknown

(i) ONE SIDED TEST

Consider a one-sided test based on 1(a) above, then we follow the following steps:

1. *Testing problem*

   $H_0: \mu = \mu_0 \qquad H_0: \mu = \mu_0$
   $H_1: \mu > \mu_0 \ \ or \ \ H_a: \mu < \mu_0$

2. *Test statistic*

   $$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

3. *Critical region*

   For H_1: reject H_0 if $z_{obs} > z_\alpha$ i.e. z_{obs} lies in the interval (z_α, ∞)

   For H_a: reject H_0 if $z_{obs} < -z_\alpha$ i.e. z_{obs} lies in the interval $(-\infty, -z_\alpha)$

4. *Observed value*

   Calculate the observed value of Z based on the data i.e. find

   $$z_{obs} = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

   Check to see if $z_{obs} > z_\alpha$ for H_1 or $z_{obs} < -z_\alpha$ for H_a

5. *Conclusion*

   Make a decision based on 3 and 4, state the conclusion.

(ii) TWO-SIDED TEST

Suppose 1(b) is the appropriate statistic, then we have the following steps:

1. *Testing problem*

   $H_0: \mu = \mu_0$
   $H_1: \mu \neq \mu_0$

2. *Test statistic*

   $$T = \frac{\bar{X} - \mu_0}{\frac{S}{\sqrt{n}}}$$

3. *Critical region*

   reject H_0 if

   $|t_{obs}| > t_{\frac{\alpha}{2}, n-1}$

4. *Observed value*

   Calculate the observed value of the test statistic T based on the data i.e. find

   $$t_{obs} = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

5. *Conclusion*

   Make a decision based on 3 and 4, state the conclusion.

NOTE

1. Sometimes z^* is used instead of z_{obs} (i.e. t^* for t_{obs})
2. If the level of significance is not stated in the question the standard level of significance of 5% is usually used.

EXAMPLES

1. A random sample of 100 deaths recorded in the United States during the past year showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, does this seem to indicate that the average life span today is greater than 70 years. Use a 0.05 level of significance.

   SOLUTION

   $$n = 100 \quad \bar{x} = 71.8 \quad \sigma = 8.9 \text{ (is known)} \quad \alpha = 0.05$$

   i.e. the sample size is large so use the test statistic $Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$

   1. *Testing problem*

      $H_0: \mu = 70$
      $H_1: \mu > 70$ } one − tailed test

2. *Test statistic*

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

3. *Critical region*

reject H_0 if $z_{obs} > z_\alpha = z_{0.05} = 1.645$

4. *Observed value*

$$z_{obs} = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{71.8 - 70}{\frac{8.9}{\sqrt{100}}} = 2.022$$

$z_{obs} = 2.022 > 1.645 = z_\alpha$

$i.e.\ \ z_{obs} > z_\alpha$

5. *Conclusion*

Reject H_0 at the 5% level of significance and conclude that the average life span is greater than 70 years.

2. The American Heart Association recommends that an individual's cholesterol level be under 200mg per 100ml. The following are readings of 16 women selected randomly from the Waterloo Heart Study:

233 197 192 179 174 217 186 221 188 209 196 167 238 179 196 191

At the 10% level of significance, do these readings suggest that women in the Waterloo area have cholesterol readings below 200 on the average? What assumption(s) are required for the conclusion to be valid?

SOLUTION

Assumptions

- The observations are a random sample (that is given in the question)
- The observations are normally distributed (this is required because the sample size is small).

The sample size is small and the population standard deviation is unknown so use a T statistic.

$$n = 16 \quad \bar{x} = 197.6875 \quad s = 20.7066 \quad \alpha = 0.1$$

1. *Testing problem*

$$\left.\begin{matrix} H_0: \mu = 200 \\ H_1: \mu < 200 \end{matrix}\right\} \text{one} - \text{tailed test}$$

2. *Test statistic*

$$T = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

3. *Critical region*

reject H_0 if $t^* < -t_{\alpha,n-1} = t_{0.10,15} = -1.341$

4. *Observed value*

$$t^* = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{197.6875 - 200}{\frac{20.7066}{4}} = -0.4467$$

$t^* = -0.4467 > -1.341 = t_{\alpha,n-1}$

i.e. $t^* > -t_{\alpha,n-1}$

5. *Conclusion*

Fail to reject H_0 at the 10% level of significance and conclude that there is insufficient evidence to conclude that the average cholesterol level of women in Waterloo is below 200.

3. A manufacturer of sports equipment has developed a new synthetic fishing line that he claims has a mean breaking strength of 8kg with a standard deviation of 0.5kg. A random sample of 50 lines is tested and found to have a mean breaking strength of 7.8kg. Is the claim valid? Use a 0.01 level of significance.

SOLUTION

$$n = 50 \quad \bar{x} = 7.8 \quad \sigma = 0.5 \quad \alpha = 0.01$$

1. *Testing problem*

$$\left.\begin{matrix} H_0: \mu = 8 \\ H_1: \mu \neq 8 \end{matrix}\right\} \text{two} - \text{tailed test}$$

2. *Test statistic*

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

3. *Critical region*

reject H_0 if $|z^*| > z_{\frac{\alpha}{2}} = z_{0.005} = 2.576$

4. *Observed value*

$$z^* = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{7.8 - 8}{\frac{0.5}{\sqrt{50}}} = -2.83$$

i.e. $|z^*| > z_{\frac{\alpha}{2}}$

5. *Conclusion*

Reject H_0 at the 0.01 level of significance. Therefore, the claim is not valid.

4. A stock market analyst claims that the average annual returns on stocks in the construction industry is 12 %. You want to test whether this claim is true. You collect the random sample of 36 stocks in the construction industry and find that the average annual return is 10% with the standard deviation of 3%. Use a 5% level of significance to test the analyst's claim.

SOLUTION

$$\left.\begin{matrix} H_0: \mu = 12\ \% \\ H_1: \mu \neq 12\% \end{matrix}\right\} \text{two} - \text{tailed test}$$

Here, the population standard deviation σ is unknown, so we use the estimate sample standard deviation, $s = 3\%$, but the sample $size\ n = 36$ is large, $\bar{x} = 10\ \%$, so we use the Z −distribution. The nature of the alternative hypothesis suggests we need to carry out a two-tailed test. Since $\alpha = 0.05$, the critical value is $\pm Z_{\frac{\alpha}{2}} = \pm Z_{0.025} = 1.96$.

The rejection criteria is therefore to reject $H_0\ if\ |Z_{cal}| > \left|\pm Z_{\frac{\alpha}{2}}\right| > 1.96$

$$Z_{cal} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{10 - 12}{3/\sqrt{36}} = 4$$

Conclusion: Since $|Z_{cal}| = 4 > 1.96$, we reject H_0 and conclude that the average annual return is not 12% and therefore, the analyst's claim is false.

5. In an advertisement it is claimed that a certain brand of air freshener will last on average at least 40 days. A random sample of 12 households took the following number of days to use up the air freshener: 28, 41, 36, 50,17, 39, 21,64,26,30,42,12

Test the claim made for the product using a 5% level of significance.

SOLUTION

$$\left.\begin{matrix} H_0: \mu \geq 40 \\ H_1: \mu < 40. \end{matrix}\right\} \text{one} - \text{tailed test}$$

The population standard deviation σ is unknown, but the sample size $n = 12$ is small so we use the t −distribution. The nature of the alternative hypothesis suggests we need to carry out a one-tailed test. Since $\alpha = 0.05$, the critical value is

$-t_{\alpha}, (n - 1) = -t_{0.05}, (11) = -1.80$, we would reject H_0 if $T_{cal} < -1.80$

You should verify that $\bar{x} = 33.8333$ and $s = 14.6339$

$$T_{cal} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$= \frac{33.8333 - 40}{14.6339/\sqrt{12}} = -0.2762$$

Conclusion: Since $T_{cal} = -0.2762 > -1.80$, we fail to reject H_0 and conclude that the data does not provide sufficient evidence at 5% level of significance to reject H_0.

(B) CONFIDENCE INTERVALS AND HYPOTHESIS TESTING (CONFIDENCE INTERVAL APPROACH TO HYPOTHESIS TESTING)

- ✓ The testing of $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$ at a significance level α is equivalent to computing a $100(1 - \alpha)\%$ confidence interval for μ and rejecting H_0 if μ_0 is not in the confidence interval.

NOTE:

1. If μ_0 is in the confidence interval, the null hypothesis $H_0: \mu = \mu_0$ is not rejected.
2. If μ_0 lies outside the confidence interval, the null hypothesis $H_0: \mu = \mu_0$ would be rejected.

- ✓ Thus, the lower and upper confidence limits form a pair of critical values beyond which H_0 will be rejected.

EXAMPLES

1. Consider the previous example 3

Testing problem

$$\left.\begin{matrix} H_0: \mu = 8 \\ H_1: \mu \neq 8 \end{matrix}\right\} \text{two} - \text{tailed test}$$

$$\alpha = 0.01$$

Then

$$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

$$7.8 \pm z_{0.005} \frac{0.5}{\sqrt{50}}$$

$$7.8 \pm 2.576 \left(\frac{0.5}{\sqrt{50}}\right)$$

$$(7.62, 7.98)$$

$\mu_0 = 8$ is not in the confidence interval, therefore reject H_0 at $\alpha = 0.01$ level of significance.

2. An electrical firm supplies light bulbs that have a length of life that is approximately normally distributed with a standard deviation of 20 hours. If a random sample of 40 bulbs has an average life of 800 hours.

(a) Find a 99% confidence interval for the population mean life of all bulbs supplied by this firm.

(b) Hence test at 1% level of significance the claim that the population mean life of all bulbs supplied by this firm is 800 hours

SOLUTION

(a) You must verify the 99% confidence interval for mean life of bulbs is found to be $(791.8546,\ 808.1454)$.

(b) the hypotheses tested are:

$H_0: \mu = 800$

$H_1: \mu \neq 800$

Assuming H_0, the claim is probably true because the confidence interval $(791.8546,\ 808.1454)$ includes 800. In other words, we fail to reject H_0.

(C) TWO POPULATION MEANS (TESTING FOR DIFFERENCE IN TWO POPULATION MEANS): INDEPENDENT RANDOM SAMPLES

✓ This is only applied for two random samples that may be independent or matched in which we want to know if there is a significant difference between their two means.

Note

1. Independent samples are those samples obtained from two populations such that the selection of one sample from a population will not affect the selection of the other sample from a different population.
2. The distribution of sample mean differences is assumed to be normally distributed.

- ✓ Let μ_1 be the population mean of the first population from which a sample is drawn and μ_2 be the mean of the second population, then the hypotheses to be tested are:

$$\text{Set A: } \begin{matrix} H_0: \mu_1 = \mu_2 \\ \mathrm{H_1}: \mu_1 \neq \mu_2 \end{matrix}$$

$$\text{Set B: } \begin{matrix} H_0: \mu_1 = \mu_2 \\ \mathrm{H_1}: \mu_1 > \mu_2 \end{matrix} \text{ or } \begin{matrix} H_0: \mu_1 \leq \mu_2 \\ \mathrm{H_1}: \mu_1 > \mu_2 \end{matrix}$$

$$\text{Set C: } \begin{matrix} H_0: \mu_1 = \mu_2 \\ \mathrm{H_1}: \mu_1 < \mu_2 \end{matrix} \text{ or } \begin{matrix} H_0: \mu_1 \geq \mu_2 \\ \mathrm{H_1}: \mu_1 < \mu_2 \end{matrix}$$

- ✓ The null hypothesis essentially says there is no difference in the means.
- ✓ Thus, the hypotheses can be expressed as differences, for example set A can be written as:

$$\begin{matrix} H_0: \mu_1 - \mu_2 = 0 \\ \mathrm{H_1}: \mu_1 - \mu_2 \neq 0 \end{matrix}$$

- ✓ The test statistic depends on whether population variances are known or not known and also on the sample size.
- ✓ We have three cases to consider:

  Case 1: when the variances (σ_1^2 and σ_2^2) are known, irrespective of the sample sizes, we use the standard normal ($Z - distribution$) distribution, and the test statistic is given by:

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

- ✓ Assuming $H_1: \mu_1 = \mu_2$, then the test statistic reduces to

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

  Case 2: when variances are unknown but samples are large.

  Sample variances s_1^2 and s_2^2 are used as estimates of population variances σ_1^2 and σ_2^2 respectively. The test statistic therefore becomes.

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim N(0,1)$$

Case 3: when variances are unknown and samples sizes are small, we use the t-distribution.

When using the $t - distribution$ we assume that

(a) the two population are normally distributed

(b) the two populations have equal variances $(\sigma_1^2 = \sigma_2^2)$ (homogeneous assumption).

(c) the two populations have no equal variances $(\sigma_1^2 \neq \sigma_2^2)$.

In this case the best estimate of the population variances in which the two populations have equal variances is the pooled sample variance given by:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n - 1)s_2^2}{n_1 + n_1 - 2}$$

Thus, the test statistic to be used will be

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{v=n_1+n_2-2}$$

when the two populations variances are not equal $(\sigma_1^2 \neq \sigma_2^2)$, the test statistic is given by

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \sim t_v$$

$$\text{where, } v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

Test statistics and critical regions for two population means can be summarised as follows:

H_0	Test statistic	H_1	Critical region
$\mu_1 - \mu_2 = \mu_0$	$Z = \dfrac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{\dfrac{\sigma_1^2}{n_1} + \dfrac{\sigma_2^2}{n_2}}}$ σ_1^2 and σ_2^2 known	$\mu_1 - \mu_2 < \mu_0$	$z^* < -z_\alpha$
		$\mu_1 - \mu_2 > \mu_0$	$z^* > z_\alpha$
		$\mu_1 - \mu_2 \neq \mu_0$	$\lvert z^* \rvert > z_{\frac{\alpha}{2}}$
$\mu_1 - \mu_2 = \mu_0$	$T = \dfrac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{s_p^2\left(\dfrac{1}{n_1} + \dfrac{1}{n_2}\right)}}$ $\nu = n_1 + n_2 - 2$ $\sigma_1^2 = \sigma_2^2$ but unknown	$\mu_1 - \mu_2 < \mu_0$	$t^* < -t_{\alpha,\nu}$
		$\mu_1 - \mu_2 > \mu_0$	$t^* > t_{\alpha,\nu}$
		$\mu_1 - \mu_2 \neq \mu_0$	$\lvert t^* \rvert > t_{\frac{\alpha}{2},\nu}$
$\mu_1 - \mu_2 = \mu_0$	$T = \dfrac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$ $\nu = \dfrac{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)^2}{\dfrac{\left(\dfrac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \dfrac{\left(\dfrac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$ $\sigma_1^2 \neq \sigma_2^2$ and unknown	$\mu_1 - \mu_2 < \mu_0$	$t^* < -t_{\alpha,\nu}$
		$\mu_1 - \mu_2 > \mu_0$	$t^* > t_{\alpha,\nu}$
		$\mu_1 - \mu_2 \neq \mu_0$	$\lvert t^* \rvert > t_{\frac{\alpha}{2},\nu}$
$\mu_D = \mu_0$	$T = \dfrac{\bar{D} - \mu_0}{\dfrac{s_d}{\sqrt{n}}}$ $\nu = n - 1$ paired observations	$\mu_D < \mu_0$	$t^* < -t_{\alpha,\nu}$
		$\mu_D > \mu_0$	$t^* > t_{\alpha,\nu}$
		$\mu_D \neq \mu_0$	$\lvert t^* \rvert > t_{\frac{\alpha}{2},\nu}$

EXAMPLES

1. An experiment was performed to compare the abrasive wear of two different laminated materials. Twelve pieces of material 1 gave an average (coded) wear of 85 units with a standard deviation of 4, while 10 pieces of material 2 gave an average of 81 and a standard

deviation of 5. Can we conclude at the 0.05 level of significance that the abrasive wear of material 1 exceeds that of material 2 by more than 2 units? Assume the populations to be normally distributed with equal variances.

SOLUTION

$$n_1 = 12 \quad \bar{x}_1 = 85 \quad s_1 = 4$$
$$n_2 = 10 \quad \bar{x}_2 = 81 \quad s_2 = 5$$
$$\alpha = 0.05$$

1. *Testing problem*

   $H_0: \mu_1 - \mu_2 = 2$
   $H_1: \mu_1 - \mu_2 > 2$

2. *Test statistic*

   $$T = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

3. *Critical region*

   reject H_0 if

   $t^* > t_{\alpha, n_1+n_2-2} = t_{0.05,20} = 1.725$

4. *Observed value*

   $$t^* = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

   $$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
   $$= \frac{11(4)^2 + 9(5)^2}{12 + 10 - 2}$$
   $$= 20.05$$

   $$t^* = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(85 - 81) - 2}{\sqrt{20.05\left(\frac{1}{12} + \frac{1}{10}\right)}} = 1.043$$

   i.e. $t^* < 1.725$

5. *Conclusion*

   We fail to reject H_0 at the 0.05 level of significance and conclude that there's insufficient evidence to conclude that the average wear for material 1 exceeds that of material 2 by more than 2 units.

2. Business schools A and B reported the following summary GMAT (Graduate Management Aptitude Test) verbal scores.

	n	$\bar{x}$	s^2
A	201	34.75	48.59
B	115	33.74	30.68

At a 5% level of significance, is there sufficient evidence to believe there is a difference in the GMAT scores of the two schools?

SOLUTION

$$n_A = 201 \quad \bar{x}_A = 34.75 \quad s_A^2 = 48.59$$
$$n_B = 115 \quad \bar{x}_B = 33.74 \quad s_A^2 = 30.68$$
$$\alpha = 0.05$$

1. *Testing problem*

   $H_0: \mu_A - \mu_B = 0$
   $H_1: \mu_A - \mu_B \neq 0$

2. *Test statistic*

$$Z = \frac{(\bar{x}_A - \bar{x}_B) - \mu_0}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

3. *Critical region*

   reject H_0 if

$$|z^*| > z_{\frac{\alpha}{2}} = z_{0.025} = 1.96$$

4. *Observed value*

$$z^* = \frac{(\bar{x}_A - \bar{x}_B) - \mu_0}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}} = \frac{(34.75 - 33.74) - 0}{\sqrt{\frac{48.59}{201} + \frac{30.68}{115}}} = 1.42$$

   i.e. $|z^*| < 1.96$

5. *Conclusion*

   Do not reject H_0 at the 0.05 level of significance. Therefore, the GMAT scores in the two schools are not significantly different.

3. A study was made to compare salaries of teachers in the private and public schools. 40 randomly selected teachers in private schools averaged $540 per month with a standard deviation of $60, while 50 randomly selected teachers in public schools averaged $360 with the standard deviation of $15. Test at 5% level of significance whether the difference between these two sample means are significant.

   SOLUTION

   The hypothesis is stated as

   $H_0: \mu_1 = \mu_2$
   $H_1: \mu_1 \neq \mu_2$
   This is a two-tailed test, variances are unknown and samples are large, therefore, we use the Z −distribution. The critical value are

   $\pm Z_{\alpha/2} = \pm Z_{0.025} = \pm 1.96$

   We would reject H_0 if $|Z_{\text{cal}}| > 1.96$, otherwise we accept H_0.

   We now calculate the test statistics:

   $$z_{cal} = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{(540 - 360) - 0}{\sqrt{\frac{60^2}{40} + \frac{15^2}{50}}} = 18.5$$

   Since $|Z_{cal}| = 18.5 > 1.96$, we reject H_0 and conclude that the salaries are different.

4. A price monitoring agency noted the price of the commodity in 8 different branches of a retail chain (A) and in 7 branches of another retail chain (B). The results were as follows:

   Retailer A: 21 19 20 21 25 23 19 22

   Retailer B: 16 20 15 18 22 20 18

   Establish at 1% level of significance whether the average price of the commodity at retailer B is significantly smaller than the price of the same commodity at retailer A.

   Solution

   The hypothesis is stated as

   $H_0: \mu_1 = \mu_2$
   $H_1: \mu_1 > \mu_2$

It is a one tailed test, variances are not known and samples are small, therefore, we use the t −distribution. Before we calculate the test statistic, we need to find the pooled sample variance.

Consider retailer A as the first sample, so that

$n_1 = 8, \quad \bar{x}_1 = 21.25, \quad s_1^2 = 4.2143$

And retailer B as the second sample to give

$n_2 = 7, \quad \bar{x}_2 = 18.4286, \quad s_2^2 = 5.9524$

Now the pooled variance is given by

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$
$$= \frac{7(4.2143) + 6(5.9524)}{13}$$
$$= 5.0165$$

Thus, the calculated test statistic value is given by

$$T_{cal} = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_0}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{(21.25 - 18.4286) - 0}{\sqrt{5.0165\left(\frac{1}{8} + \frac{1}{7}\right)}} = 0.7935$$

The critical value is given by

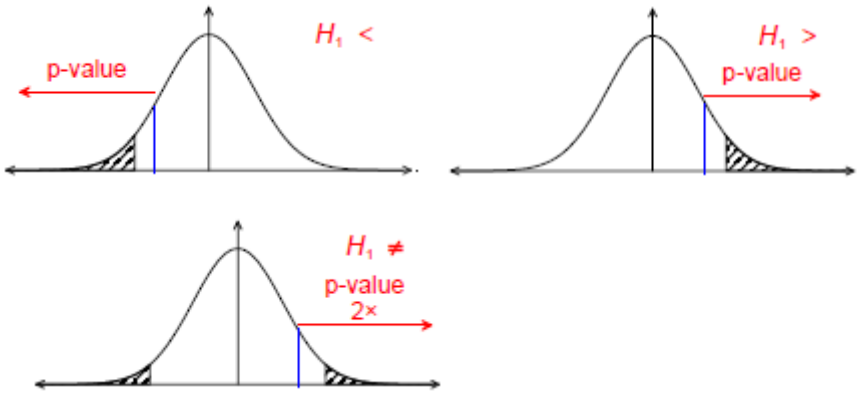
$$t_\alpha(n_1 + n_2 - 2) = t_{0.01}(8 + 7 - 2) = t_{0.01}(13) = 2.65$$

Decision and conclusion: We would reject H_0 if $T_{cal} > 2.65$. Thus, since $T_{\text{cal}} = 0.7935 < 2.65$, we do not reject H_0 and conclude that there is insufficient evidence that the average price of the commodity is smaller at retailer B

3.8. P-VALUE OF A STATISTICAL TEST

- ✓ Definition ($p - value$): The p − value of a statistical test is the probability of getting sample results as extreme as the results that were actually obtained when assuming the null hypothesis is true.
- ✓ In other words, the p −value is the probability of observing a test statistic as extreme as or more extreme than the observed value, if in fact H_0 is true.
- ✓ We define “extreme” as far below or far above what we would have expected.
- ✓ In general, if we assume that the null hypothesis is true, the *p*-value can be defined as the probability that a sample statistic (such as the sample mean) is at least as far away from the hypothesized value of the parameter in the direction of the alternative

hypothesis as the one obtained from the sample data under consideration as shown in the diagrams below:



NOTE:

1. The p −value or observed significance level of a statistical test is the smallest significance level or smallest value at which the null hypothesis is rejected.
2. The p −value is the actual risk of committing a type I error, if H_0 is rejected based on the observed value of the test statistic.
3. The p −value measures the strength of the evidence against H_0.
4. The p −value is a probability calculated using the test statistic
5. For a one-tailed test, the p −value is given by the area in the tail of the sampling distribution curve beyond the observed value of the sample statistic,
6. For a right-tailed test, the p −value about μ will be the area in the upper tail of the sampling distribution curve to theright of the observed value of $\bar{x}$
7. For a left-tailed test, the p −value will be the area in the lower tail of the sampling distribution curve to the left of the observed value of $\bar{x}$.
8. For a two-tailed test, the p −value is actually twice the area in the tail of the sampling distribution curve beyond the observed value of the sample statistic or twice the tail area corresponding to the calculated value of the test statistic.

(A) REJECTION AND ACCEPTANCE REGION FOR *P* −VALUE

1. We reject the null hypothesis and report that the results are statistically significant at level α if the p −value is less than or equal to a pre-assigned significance level α, that is, $p-\text{value} \leq \alpha$ or $\alpha \geq p-\text{value}$, where α can be 1% or 5% etc.

2. We do not reject the null hypothesis the $p-$value is greater than a pre-assigned significance level α, that is, $p-\text{value} > \alpha$ or $\alpha < p-\text{value}$

(B) STEPS TO PERFORM A TEST OF HYPOTHESIS USING THE *P* −VALUE APPROACH

1. State the null and alternative hypotheses.
2. Select the distribution to use.
3. Calculate the $p-$value.
4. Make a decision.

(C) The *P* −VALUE APPROACH FOR CASE 1 (a) AND CASE 2 (a) AND (b)

To find the area under the normal distribution curve beyond the sample mean $\bar{x}$, we first find the Zvalue for $\bar{x}$ using the following formula"

$$Z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}, \text{where } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

The value of Z calculated for $\bar{x}$ using this formula is also called the observed value of Z.

EXAMPLES

1. At Canon Food Corporation, it used to take an average of 90 minutes for new workers to learn a food processing job. Recently the company installed a new food processing machine. The supervisor at the company wants to find if the meantime taken by new workers to learn the food processing procedure on this new machine is different from 90 minutes. A random sample of 20 workers showed that it took, on average, 85 minutes for them to learn the food processing procedure on the new machine. It is known that the learning times for all new workers are approximately normally distributed with a population standard deviation of 7 minutes. Find the $p-$value for the test that the mean learning time for the food processing procedure on the new machine is different from 90 minutes. What will your conclusion be if $\alpha = 0.01$?

SOLUTION

Let μ be the mean time (in minutes) taken to learn the food processing procedure on the new machine by all workers, and let $\bar{X}$ be the corresponding sample mean. From the given information,

$$n = 20, \bar{x} = 85\ minutes, \qquad \sigma = 7\ minutes, \qquad \text{and } \alpha = 0.01$$

To calculate the p −value and perform the test, we apply the following four steps.

Step 1. *State the null and alternative hypotheses*.

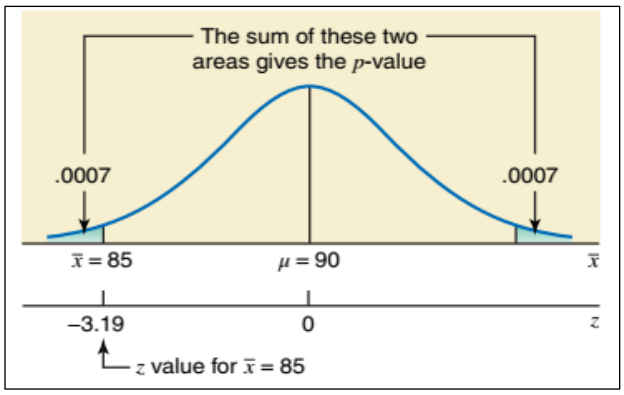
$$H_0: \mu = 90 \text{ minutes}$$
$$H_1: \mu \neq 90 \text{ minutes}$$

Step 2. *Select the distribution to use*.

Here, the population standard deviation σ is known, the sample size is small $(n < 30)$, but the population distribution is approximately normal. Hence, the sampling distribution of $\bar{X}$ is approximately normal with its mean equal to μ and the standard deviation equal to $\sigma_{\bar{X}} = \sigma / \sqrt{n}$. Consequently, we will use the z −normal distribution to find the p −value and make the test.

Step 3. *Calculate the p-value.*

The ≠ sign in the alternative hypothesis indicates that the test is two-tailed. The p −value is equal to twice the area in the tail of the sampling distribution curve of $\bar{x}$ to the left of $\bar{x} = 85$, as shown in the diagram below.



To find this area, we first find the Z value for $\bar{x} = 85$ as follows:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{7}{\sqrt{20}} = 1.56524758 \text{ minutes}$$

$$Z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{85 - 90}{1.56524758} = -3.19$$

The area to the left of $\bar{x} = 85$ is equal to the area under the standard normal curve to the left of $Z = -3.19$. From the normal distribution table, the area to the left of $Z = -3.19$ is .0007.

Consequently, the p −value is

$$P(z < -3.19) + P(z > 3.19) = 2 \cdot P(z < -3.19)$$
$$= 2(0.0007) = 0.0014$$

Step 4. *Make a decision.*

Thus, based on the *p*-value of .0014, we can state that for any α (significance level) greater than or equal to .0014, we will reject the null hypothesis stated in Step 1, and for any α less than .0014, we will not reject the null hypothesis. Because $\alpha = 0.01$ is greater than the *p*-value of .0014, we reject the null hypothesis at this significance level. Therefore, we conclude that the mean time for learning the food processing procedure on the new machine is different from 90 minutes.

2. The management of Priority Health Club claims that its members lose an average of 10 pounds or more within the first month after joining the club. A consumer agency that wanted to check this claim took a random sample of 36 members of this health club and found that they lost an average of 9.2 pounds within the first month of membership. The population standard deviation is known to be 2.4 pounds. Find the *p*-value for this test. What will your decision be if $\alpha = 0.01$? What if $\alpha = 0.05$?

SOLUTION

Let μ be the mean weight lost during the first month of memberships by all members of this health club, and let $\bar{X}$ be the corresponding sample mean. From the given information,

$n = 36, \quad \bar{x} = 9.2\, pounds, and\ \sigma = 2.4\, pounds,$ and $\alpha = 0.05$

To calculate the p −value and perform the test, we apply the following four steps.

Step 1. *State the null and alternative hypotheses*.

$H_0: \mu \geq 10$ (The mean weight lost is 10 pounds or more.)
$H_1: \mu < 10$ (The mean weight lost is less than 10 pounds)

Step 2. *Select the distribution to use*.

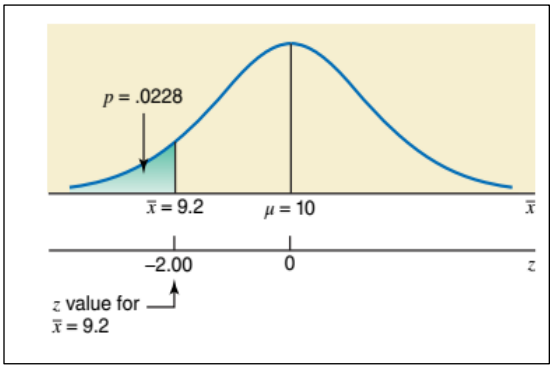
Here, the population standard deviation σ is known, and the sample size is large ($n \geq 30$), but the population distribution is approximately normal. Hence, the sampling distribution of $\bar{x}$ is normal (due to the central limit theorem) with its mean

equal to μ and the standard deviation equal to $\sigma_{\bar{x}} = \sigma / \sqrt{n}$. Consequently, we will use the normal distribution to find the p −value and perform the test.

Step 3. *Calculate the p-value.*

The $<$ sign in the alternative hypothesis indicates that the test is left-tailed.

The p −value is given by the area to the left of $\bar{x} = 9.2$ under the sampling distribution curve of $\bar{x}$ as shown in the diagram below.



To find this area, we first find the Z value for $\bar{x} = 9.2$ as follows:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{2.4}{\sqrt{36}} = 0.40$$

$$Z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{9.2 - 10}{0.40} = -2.00$$

The area to the left of $\bar{x} = 9.2$ under the sampling distribution of $\bar{x}$ is equal to the area under the standard normal curve to the left of $Z = -2.00$. From the normal distribution table, the area to the left of $Z = -2.00$ is 0.0228. Consequently, the p −value is

$$P(z < -2.00) = 0.0228$$

Step 4. *Make a decision.*

Thus, based on the $p - value$ of .0228, we can state that for any α (significance level) greater than or equal to .0228, we will reject the null hypothesis stated in Step 1, and for any α less than .0228, we do not will not reject the null hypothesis. Since $\alpha = 0$ .01 is less than the *p*-value of .0228, we do not reject the null hypothesis at this significance level. Consequently, we conclude that there is no significant evidence to reject the claim that the mean weight lost within the first month of membership by the members of this club is 10 pounds or more. Now

because $\alpha = 0.05$ is greater than the p-value of .0228, we reject the null hypothesis at this significance level. In this case we conclude that the mean weight lost within the first month of membership by the members of this club is less than 10 pounds.

(D) THE P −VALUE APPROACH FOR CASE 1 (b)

- ✓ For case 1 (b) above, we use the $t - distribution$ perform a test of hypothesis about μ.
- ✓ Similarly, to use the $p - value$ approach we will use the $t - distribution$ to perform a test of hypothesis about μ.
- ✓ To do this, we will use the same four steps that we used in performing a test of hypothesis for normal distribution.

NOTE

when using the $t - distribution$ and $t - table$, we cannot find the exact $p - value$ but only a range within which it falls.

EXAMPLES

The following examples illustrate the p −value procedure to test a hypothesis about μ using the t distribution.

1. A psychologist claims that the mean age at which children start walking is 12.5 months. Carol wanted to check if this claim is true. She took a random sample of 18 children and found that the mean age at which these children started walking was 12.9 months with a standard deviation of 0.80 month. It is known that the ages at which all children start walking are approximately normally distributed. Find the p −value for the test that the mean age at which all children start walking is different from 12.5 months. What will your conclusion be if the significance level is 1%?

   SOLUTION

   Let μ be the mean age at which all children start walking, and let $\bar{x}$ be the corresponding mean for the sample. Then, from the given information,

   $n = 18, \bar{x} = 12.9$ months, $s = 0.80$ month, and $\alpha = 0.01$

   Step 1. *State the null and alternative hypotheses*.

   We are to test if the mean age at which all children start walking is different from 12.5 months.

The null and alternative hypotheses are:

$H_0: \mu = 12.5$ (The mean walking age is 12.5 months.)

$H_1: \mu \neq 12.5$ (The mean walking age is different from 12.5 months.)

Step 2. *Select the distribution to use*.

In this example, the population standard deviation σ is not known, the sample size is small ($n < 30$), and the population is approximately normally distributed. Hence, we will use the *t* distribution to perform the test in this example.

Step 3. *Calculate the p-value.*

The ≠ sign in the alternative hypothesis indicates that the test is two-tailed. To find the *p*-value, first we find the degrees of freedom and the t value for $x = 12.9$ months. Then, $p-$value is equal to twice the area in the tail of the $t - distribution$ curve beyond this *t* value for $\bar{X} = 12.9$ months**.** We find this *p*-value as follows:

$$s_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{0.80}{\sqrt{18}} = 0.18856181$$

$$T = \frac{\bar{x} - \mu}{S_{\bar{x}}} = \frac{12.9 - 12.5}{0.18856181} = 2.121$$

and $df = n - 1 = 18 - 1 = 17$

Now we can find the range for the *p*-value. To do so, we go to the *t* distribution table and find the row of $df = 17$. In this row, we find the two values of t that cover $t = 2.121$. From t table, for $df = 17$, these two values of t are 2.110 and 2.567.

The test statistic $t = 2.121$ falls between these two values. Now look in the top row of this table to find the areas in the tail of the t distribution curve that correspond to 2.110 and 2.567. These two areas are .025 and .01, respectively. In other words, the area in the upper tail of the *t* distribution curve for $df = 17$ and $t = 2.110$ is .025, and the area in the upper tail of the *t* distribution curve for $df = 17$ and $t = 2.567$ is .01. Because it is a two-tailed test, the *p*-value for $t = 2.121$ is between 2(.025) = .05 and 2(.01) = .02, which can be written as.

$$02 < p - value < .05$$

Step 4. *Make a decision.*

Thus, we can state that for any α greater than or equal to 0.05 (the upper limit of the *p*-value range), we will reject the null hypothesis and for any α less than or equal to .02 (the lower limit of the *p*-value range), we will not reject the null hypothesis.

However, if α is between .02 and .05, we cannot make a decision. Hence, since $\alpha = 0.01$, which is less than the lower limit of the *p*-value range of .02, we fail to reject H_0 and conclude that the mean age at which all children start walking is not significantly different from 12.5 months.

3.9. TESTS ABOUT PROPORTIONS

- ✓ Tests about population proportions are based on the sampling distribution of proportion for a single population $(\hat{P})$ or for difference of two population proportions $(\hat{P}_1 - \hat{P}_2)$.
- ✓ Here we test the hypothesis that the proportions in two populations are not different.
- ✓ The notation to be used is summarized as follows:

	Sample 1	Sample 2
Sample size	n_1	n_2
Sample proportion	$\hat{P}_1$	$\hat{P}_2$
Population proportion	p_1	p_2

- ✓ The hypotheses to be tested are

$$\text{Set A: } \begin{matrix} H_0: p_1 = p_2 \\ \mathrm{H_1}: \mathrm{p_1} \neq \mathrm{p_2} \end{matrix}$$

$$\text{Set B: } \begin{matrix} H_0: p_1 = p_2 \\ \mathrm{H_1}: \mathrm{p_1} > \mathrm{p_2} \end{matrix} \text{ or } \begin{matrix} H_0: p_1 \leq p_2 \\ \mathrm{H_1}: \mathrm{p_1} > \mathrm{p_2} \end{matrix}$$

$$\text{Set C: } \begin{matrix} H_0: p_1 = p_2 \\ \mathrm{H_1}: \mathrm{p_1} < \mathrm{p_2} \end{matrix} \text{ or } \begin{matrix} H_0: p_1 \geq p_2 \\ \mathrm{H_1}: \mathrm{p_1} < \mathrm{p_2} \end{matrix}$$

- ✓ The test statistics and critical regions are given below:

H_0	Test statistic	H_1	Critical region
$p = p_0$	$Z = \dfrac{\hat{P} - p_0}{\sqrt{\dfrac{p_0(1-p_0)}{n}}}$	$p < p_0$	$z^* < -z_\alpha$
		$p > p_0$	$z^* > z_\alpha$
		$p \neq p_0$	$\lvert z^* \rvert > z_{\frac{\alpha}{2}}$
$p_1 - p_2 = 0$	$Z = \dfrac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{P}(1-\hat{P})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$ where $\hat{P} = \dfrac{X_1 + X_2}{n_1 + n_2}$	$p_1 - p_2 < 0$	$z^* < -z_\alpha$
		$p_1 - p_2 > 0$	$z^* > z_\alpha$
		$p_1 - p_2 \neq 0$	$\lvert z^* \rvert > z_{\frac{\alpha}{2}}$
$p_1 - p_2 = p_0$	$Z = \dfrac{\hat{P}_1 - \hat{P}_2 - p_0}{\sqrt{\dfrac{\hat{P}_1(1-\hat{P}_1)}{n_1} + \dfrac{\hat{P}_2(1-\hat{P}_2)}{n_2}}}$	$p_1 - p_2 < p_0$	$z^* < -z_\alpha$
		$p_1 - p_2 > p_0$	$z^* > z_\alpha$
		$p_1 - p_2 \neq p_0$	$\lvert z^* \rvert > z_{\frac{\alpha}{2}}$

NOTE

1. There is a difference in the standard errors used in confidence intervals and hypothesis testing e.g. for a single proportion we use the standard error $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ for the confidence interval and $\sqrt{\frac{p_0(1-p_0)}{n}}$ for testing hypothesis.
2. In hypothesis testing the test statistic assumes the null hypothesis H_0 is true, that is why we use p_0 instead of $\hat{p}$ in the standard error for a single proportion. Similarly, to test $H_0: p_1 - p_2 = 0$ we assume the null hypthesis is true and estimate a common estimate of p $\left(\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}\right)$. If $H_0: p_1 - p_2 = p_0$ then under H_0 we have different proportions and hence the standard error uses different estimates of the two proportions.

EXAMPLES

1. A poll is taken to compare the proportion of town and country voters favouring a proposal of constructing a chemical plant. Is the proportion of town voters favouring the proposal higher than the proportion of country voters, if 120 of 200 town voters and 240 of 500 country residents favour the proposal? Use a 0.025 level of significance.

   SOLUTION

   Let $p_1 = proportion\ of\ town\ voters\ favouring\ the\ proposal$

   $p_2 = proportion\ of\ country\ voters\ favouring\ the\ proposal$

   $$n_1 = 200 \quad x_1 = 120 \quad \hat{p}_1 = \frac{x_1}{n_1} = \frac{120}{200} = 0.6$$

   $$n_2 = 500 \quad x_2 = 240 \quad \hat{p}_2 = \frac{x_2}{n_2} = \frac{240}{500} = 0.48$$

   $$\alpha = 0.025$$

   1. *Testing problem*

      $H_0: p_1 = p_2\ (p_1 - p_2 = 0)$
      $H_1: p_1 > p_2\ (p_1 - p_2 > 0)$

   2. *Test statistic*

      $$Z = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{P}(1-\hat{P})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

   3. *Critical region*

      reject H_0 if

      $z^* > z_\alpha = z_{0.025} = 1.96$

   4. *Observed value*

      $$z^* = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{120 + 240}{200 + 500} = \frac{360}{700} = 0.51$$

      $$z^* = \frac{0.6 - 0.48}{\sqrt{0.51(1-0.51)\left(\frac{1}{200} + \frac{1}{500}\right)}} = 2.87$$

      i.e. $z^* > 1.96$

5. *Conclusion*

   Reject H_0 at the 0.025 level of significance. Therefore, the proportion of town voters favouring the proposal is higher.

2. A distributor of washing powder claims that 20% of women in Lusaka prefer Boom washing powder. To test this claim, 20 women are selected at random and asked what brand they prefer. If 6 out of 20 name Boom as their preference, what conclusion do we draw? Use a 0.05 level of significance.

   SOLUTION

$$n = 20 \qquad x = 6 \qquad \hat{p} = \frac{x}{n} = \frac{6}{20} = 0.3$$

$$\alpha = 0.05$$

1. *Testing problem*

   $H_0: p = 0.20$
   $H_1: p \neq 0.20$

2. *Test statistic*

$$Z = \frac{\hat{P} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

3. *Critical region*

   reject H_0 if

   $|z^*| > z_{\frac{\alpha}{2}} = z_{0.025} = 1.96$

4. *Observed value*

$$z^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.3 - 0.2}{\sqrt{\frac{0.2(1-0.2)}{20}}} = 1.12$$

   i.e. $|z^*| < 1.96$

5. *Conclusion*

   Do not reject H_0 at the 0.05 level of significance. Therefore, the claim is valid.

3.10. TESTS ABOUT VARIANCES

The test statistics and critical regions concerning variances can be summarised as follows:

H_0	Test statistic	H_1	Critical region
$\sigma^2 = \sigma_0^2$	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$	$\sigma^2 > \sigma_0^2$	$\chi^2_{obs} > \chi^2_{\alpha,n-1}$
		$\sigma^2 < \sigma_0^2$	$\chi^2_{obs} < \chi^2_{1-\alpha,n-1}$
		$\sigma^2 \neq \sigma_0^2$	$\chi^2_{obs} < \chi^2_{1-\frac{\alpha}{2},n-1}$ or $\chi^2_{obs} > \chi^2_{\frac{\alpha}{2},n-1}$
$\sigma_1^2 = \sigma_2^2$	$F = \frac{s_1^2}{s_2^2}$	$\sigma_1^2 > \sigma_2^2$	$f^* > f_\alpha(n_1 - 1, n_2 - 1)$
		$\sigma_1^2 < \sigma_2^2$	$f^* < f_{1-\alpha}(n_1 - 1, n_2 - 1)$
		$\sigma_1^2 \neq \sigma_2^2$	$f^* < f_{1-\frac{\alpha}{2}}(n_1 - 1, n_2 - 1)$ or $f^* > f_{\frac{\alpha}{2}}(n_1 - 1, n_2 - 1)$

EXAMPLES

1. The caffeine content of a certain brand of tea is known to be normally distributed with a variance of 1.3 mg. Test the hypothesis that $\sigma^2 = 1.3$ against the alternative that $\sigma^2 \neq 1.3$ if a random sample of 8 packets of tea has a standard deviation of 1.8. Use a 0.05 level of significance.

   Solution

   $$n = 8 \quad s = 1.8 \quad \alpha = 0.05$$

   1. *Testing problem*

      $H_0: \sigma^2 = 1.3$
      $H_1: \sigma^2 \neq 1.3$

   2. *Test statistic*

      $$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$$

   3. *Critical region*

reject H_0 if

$$\chi^2_{obs} < \chi^2_{1-\frac{\alpha}{2},n-1} = \chi^2_{0.975,7} = 1.69 \; or \; \chi^2_{obs} > \chi^2_{\frac{\alpha}{2},n-1} = \chi^2_{0.025,7} = 16.013$$

4. *Calculation (Observed value)*

$$\chi^2_{obs} = \frac{(n-1)s^2}{\sigma_0^2} = \frac{7(1.8^2)}{1.3} = 17.446$$

i.e. $\chi^2_{obs} > 16.013$

5. *Conclusion*

Reject H_0 at the 0.05 level of significance and conclude that $\sigma^2 \neq 1.3$.

2. In the abrasive wear example, we assumed that the two unknown population variances were equal. Were we justified in making the assumption? Use a 0.10 level of significance.

$Material\ 1$: $n_1 = 12 \quad \bar{x}_1 = 85 \quad s_1 = 4$

$Material\ 2$: $n_2 = 10 \quad \bar{x}_2 = 81 \quad s_2 = 5$

SOLUTION

$\alpha = 0.1$

1. *Testing problem*

$H_0: \sigma_1^2 = \sigma_2^2$
$H_1: \sigma_1^2 \neq \sigma_2^2$

2. *Test statistic*

$$F = \frac{s_1^2}{s_2^2}$$

3. *Critical region*

reject H_0 if

$$f^* < f_{1-\frac{\alpha}{2}}(n_1 - 1, n_2 - 1) \quad or \; f^* > f_{\frac{\alpha}{2}}(n_1 - 1, n_2 - 1)$$
$$f^* < f_{0.95}(11,9) \quad or \; f^* > f_{0.05}(11,9)$$
$$f^* < \frac{1}{f_{0.05}(9,11)} \quad or \; f^* > \frac{3.14 + 3.07}{2}$$
$$f^* < \frac{1}{2.90} \quad or \; f^* > 3.105$$
$$f^* < 0.345$$

Note that we do not have 11 degrees of freedom on our F table, so we use the average of 10 and 12 numerator degrees of freedom.

4. *Calculation (Observed value)*

$$f^* = \frac{s_1^2}{s_2^2} = \frac{4^2}{5^2} = 0.64$$

i.e. 0 $0.345 < f^* < 3.105$

5. *Conclusion*

Do not reject H_0 at the 0.1 level of significance. The variances are not significantly different. Therefore, we were justified in assuming that the variances were equal.

3.11. TESTS FOR GOODNESS-OF-FIT AND INDEPENDENCE

These tests are based on the chi-square distribution.

(A) GOODNESS-OF-FIT TEST

- ✓ This test is based on how good a fit we have between the frequency of occurrence of observations in an observed sample and the expected frequencies obtained from a hypothesized distribution.
- ✓ **THEOREM:** A goodness of fit test between observed and expected frequencies is based on

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

where χ^2 has an approximate chi-square distribution with $k - 1$ degrees of freedom with O_i and E_i being observed and expected frequencies for the ith cell.

NOTE:

1. If the difference between the observed and expected frequencies is small then χ^2_{obs} will be small.
2. If the difference is large then χ^2_{obs} is large and the hypothesis of the hypothesized distribution will be rejected.

EXAMPLE

A die is thrown 120 times and the number of times each outcome appears is recorded. The following results are obtained:

Outcome	1	2	3	4	5	6
Observed	20	22	17	18	19	24

Is the die balanced? Use $\alpha = 0.05$.

SOLUTION

$$\alpha = 0.05$$

1. *Testing problem*

   H_0: *the die is balanced*
   H_1: *the die is not balanced*

2. *Test statistic*

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

3. *Critical region*

   reject H_0 if

$$\chi^2_{obs} > \chi^2_{\alpha,k-1} = \chi^2_{0.05,5} = 11.071$$

4. *Calculation*

Outcome	1	2	3	4	5	6
Observed	20	22	17	18	19	24
Expected	20	20	20	20	20	20

Note that if the die is balanced then we expect each outcome to appear the same number of times i.e. $\frac{120}{6} = 20$

$$\chi^2_{obs} = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = \frac{(20-20)^2}{20} + \frac{(22-20)^2}{20} + \frac{(17-20)^2}{20} + \frac{(18-20)^2}{20} + \frac{(19-20)^2}{20} + \frac{(24-20)^2}{20}$$
$$= 1.7$$

i.e. $\chi^2_{obs} < 11.071$

5. *Conclusion*

   Fail to reject H_0 at the 0.05 level of significance. Therefore, the die is balanced.

(B) TEST FOR INDEPENDENCE

Suppose a random sample is cross classified based on two variables and the observed frequencies recorded as follows:

		Variable 2				
		1	2		c	Total
Variable 1	1	O_{11}	O_{12}	$\cdots$	O_{1c}	$O_{1\bullet}$
	2	O_{21}	O_{22}	$\cdots$	O_{2c}	$O_{2\bullet}$
	$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
	r	O_{r1}	O_{r2}	$\cdots$	O_{rc}	$O_{r\bullet}$
Total		$O_{\bullet 1}$	$O_{\bullet 2}$	$\cdots$	$O_{\bullet c}$	$O_{\bullet\bullet}$

Grand total

We can test whether variable 1 and variable 2 are independent by comparing the observed and expected frequencies and use the χ^2 statistic

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)}$$

where

$$E_{ij} = \frac{O_{i\bullet} \times O_{\bullet j}}{O_{\bullet\bullet}}$$
$$= \frac{(row\ total) \times (column\ total)}{grand\ total}$$

We reject the null hypothesis of independence if χ^2_{obs} is large $\left(\chi^2_{obs} > \chi^2_{\alpha,(r-1)(c-1)}\right)$. χ^2 is an approximate χ^2 distribution with $\nu = (r-1)(c-1)$ degrees of freedom.

EXAMPLE

In an experiment to study the dependence of hypertension on smoking habits, the following data were collected on 180 individuals

	Nonsmokers	Moderate smokers	Heavy smokers
Hypertension	21	36	30
No hypertension	48	26	19

Test the hypothesis that presence or absence of hypertension is independent of smoking habits. Use a 0.05 level of significance.

SOLUTION

1. *Testing problem*

   H_0: *smoking and hypertension are independent*
   H_1: *smoking and hypertension are not independent*

2. *Test statistic*

$$\chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

3. *Critical region*

   reject H_0 if

$$\chi^2_{obs} > \chi^2_{\alpha,(r-1)(c-1)} = \chi^2_{0.05,(2-1)(3-1)} = \chi^2_{0.05,2} = 5.991$$

4. *Calculation*

$$\chi^2_{obs} = \sum_{i=1}^{2}\sum_{j=1}^{3}\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

		1	2	3	
		N	MS	HS	Total
1	H	21 $E_{11} = 33.35$	36 $E_{12} = 29.97$	30 23.68	87
2	N H	48 35.65	26 32.03	19 25.32	93
	Total				

$$E_{11} = \frac{O_{1\bullet} \times O_{\bullet 1}}{O_{\bullet\bullet}} = \frac{87 \times 69}{180} = 33.35$$

$$E_{12} = \frac{O_{1\bullet} \times O_{\bullet 2}}{O_{\bullet\bullet}} = \frac{87 \times 62}{180} = 29.97$$

$$\vdots$$

And so on you, can complete the table….

$$\chi^2_{obs} = \frac{(21-33.35)^2}{33.35} + \frac{(36-29.97)^2}{29.97} + \frac{(30-23.68)^2}{23.68} + \frac{(48-35.65)^2}{35.65} + \frac{(26-32.03)^2}{32.03} + \frac{(19-25.32)^2}{25.32}$$
$$= 14.46$$

i.e. $\chi^2_{obs} > 5.991$

5. *Conclusion*

   Reject H_0 at the 0.05 level of significance. Therefore, smoking and hypertension are not independent.

4. ANALYSIS OF VARIANCE (ANOVA)

4.1. INTRODUCTION

- ✓ So far we have looked at methods or techniques for comparing two population means or how to compare the population means when the data came from one or two populations.
- ✓ We now look at a method for comparing several population means at the same time which is called the analysis of variance.
- ✓ The short form for analysis of variance is ANOVA. ANOVA is a procedure that is used to test the null hypothesis that the means of three or more populations are all equal.
- ✓ In other words, ANOVA is one of the most flexible and practical techniques for comparing several population means or the analysis of variance is applied to the inferential analysis of means across groups or treatments.
- ✓ In general, in the analysis of variance (ANOVA) we are concerned with studying situations involving the comparison of more than two population or treatment means.

4.2 THE ANOVA TEST

- ✓ As earlier stated, the ANOVA is designed to test the difference between means of more than two samples from normal populations with the same variance.
- ✓ In ANOVA, we introduce some tests that are used to analyze the data from more than two populations.
- ✓ These tests are used to deal with treatment effects, including tests that take into account other factors that may affect the response.
- ✓ The hypothesis that the population means are equal is considered equivalent to the hypothesis that there is no difference in treatment effects.

- ✓ The analytical method we will use in such problems is called the analysis of variance (ANOVA).
- ✓ Now suppose we have k populations of interest. The method of analysis of variance (ANOVA) enables us (under suitable conditions) to test the hypothesis

  $H_0: \mu_1 = \mu_2 = \cdots = \mu_k$ (all the means are equal difference between the means)
  $H_1: not\ all\ the\ means\ are\ equal$
- ✓ Alternatively, we can set the hypotheses as follows:

  H_0: there is no significant difference between the means
  H_1: there is a significant difference between the means
- ✓ An *analysis of variance* test is performed using the ***F distribution***.
- ✓ Like the chi-square distribution, the shape of a particular ***F −distribution*** curve depends on the number of degrees of freedom.
- ✓ However, the *F* distribution has *two* numbers of degrees of freedom, that is ***degrees of freedom for the numerator*** and ***degrees of freedom for the denominator***.
- ✓ These two numbers representing two types of degrees of freedom are the ***parameters of the F distribution.***
- ✓ Each combination of degrees of freedom for the numerator and for the denominator gives a different *F* distribution curve.
- ✓ The values of an *F* distribution are denoted by *F*, which assumes only non-negative values.
- ✓ Like the normal, *t*, and chi-square distributions, the *F* distribution is a continuous distribution.
- ✓ The shape of an *F* distribution curve is skewed to the right, but the skewness decreases as both numbers of degrees of freedom increase.

NOTE

1. ANOVA is not about analysing the population variance, it is about analysing treatment means by identifying sources of variability of the data.
2. In its simplest form, ANOVA can be considered as an extension of the test of hypothesis for the equality of two means that we learned in the previous session of this unit.

4.3 ONE WAY ANALYSIS OF VARIANCE (COMPLETELY RANDOMIZED DESIGN)

- ✓ The one-way Analysis of Variance (ANOVA) is a generalization of the two-means procedure to a test of equality of the means of more than two independent, normally distributed populations.
- ✓ It is refers to a situation where there is one factor with a number of levels e.g. measure baking at 3 temperature levels (low, medium, high), in this case the factor is temperature and the levels are low, medium and high.
- ✓ The levels of a factor are often called treatments.

NOTE: For the ANOVA procedure to be valid we must either

1. pick random samples from k populations or
2. randomly assign subjects or units to k treatments (this balances out known and unknown factors).

(A) NOTATION

- ✓ The observations from the k treatments can be represented in a table below where y_{ij} represents the jth observation from the ith treatment.

	Treatment				
Observations	1	2	...	k	Total
1	y_{11}	y_{21}	...	y_{k1}	$y_{\bullet 1}$
2	y_{12}	y_{22}	...	y_{k2}	$y_{\bullet 2}$
⋮	⋮	⋮		⋮	⋮
y_n	y_{1n}	y_{2n}	...	y_{kn}	$y_{\bullet k}$
Total	$y_{1\bullet}$	$y_{2\bullet}$	...	$y_{k\bullet}$	$y_{\bullet\bullet}$

Where,

$$y_{i\bullet} = \sum_{j=1}^{n} y_{ij} \quad , \quad \bar{y}_{i\bullet} = \frac{y_{i\bullet}}{n} \ \ (mean\ for\ treatment\ i)$$

$$y_{\bullet\bullet} = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij} \quad , \quad \bar{y}_{\bullet\bullet} = \frac{y_{\bullet\bullet}}{nk} \ \ (overall\ mean\ )$$

(B) ASSUMPTIONS OF ONE-WAY ANOVA

- ✓ The following assumptions must hold true to use one-way ANOVA.

1. The populations from which the samples are drawn are approximately normally distributed.
2. The populations from which the samples are drawn have the same or common variance (or standard deviation).
3. The samples or observations drawn from different populations are random and independent.

- ✓ Thus, to test the hypothesis

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_k$$
$$H_1: not\ all\ the\ means\ are\ equal$$

- ✓ Here, we assume that the k populations are independent and normally distributed with means $\mu_1,\ \mu_2, \cdots, \mu_k$ and a common variance σ^2. This can be written as a model as

$$Y_{ij} = \mu_i + \varepsilon_{ij}\ , i = 1, 2, \ldots, k$$
$$j = 1, 2, \ldots, n$$

where

$Y_{ij} = j^{th}\ response\ from\ i^{th}\ population$

$\mu_i = mean\ of\ the\ i^{th}\ population$

$\varepsilon_{ij} = random\ (independent)\ error$ and

$\varepsilon_{ij} \sim N(0, \sigma^2)$

(C) ANOVA TABLE

- ✓ To test the hypothesis

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_k$$
$$H_1: not\ all\ the\ means\ are\ equal$$

- ✓ we construct an ANOVA table as follows:

Source (SOV)	(SS)	(df)	(MS)	F^*
Treatments	$SSTrt$	$k-1$	$MSTrt = \frac{SSTrt}{k-1}$	$\frac{MSTrt}{MSE}$
Error	SSE	$N-k$	$MSE = \frac{SSE}{N-k}$	
Total	SST	$N-1$		

Where,

$SSTrt = Treatment\ sum\ of\ squares$
$SSE = Error\ sum\ of\ squares$
$SST = Total\ sum\ of\ squares$
$MSTrt = Mean\ treatment\ sum\ of\ squares$
$MSE = Mean\ square\ error$

$$SST = \sum_{i=1}^{k}\sum_{j=1}^{n}\left(y_{ij} - \bar{y}_{\bullet\bullet}\right)^2$$
$$= \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{\bullet\bullet}^2}{N}$$
$$SSTrt = \sum_{i=1}^{k}\sum_{j=1}^{n}(\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2$$
$$= \sum_{i=1}^{k}\frac{y_{i\bullet}^2}{n} - \frac{y_{\bullet\bullet}^2}{N}$$
$$SSE = SST - SSTrt$$

where $N = nk$ is the total number of observations.

We reject H_0 if $F^* > F_\alpha(k-1, N-k)$.

EXAMPLE

A psychologist was interested in the effects of 3 kinds of drugs on the mean time to complete a task. She randomly assigned 5 subjects to each drug A, B and C and measured the time (in minutes) to complete the task. Are the population mean times the same for each drug?

Drug		
A	B	C
20	21	30
22	26	24
25	26	26
24	27	25
19	25	30

SOLUTION

Let

$\mu_1 = mean\ time\ for\ drug\ A$
$\mu_2 = mean\ time\ for\ drug\ B$
$\mu_3 = mean\ time\ for\ drug\ C$

1. *Testing problem*

   $H_0: \mu_1 = \mu_2 = \mu_3$
   $H_1: not\ all\ the\ 3\ means\ are\ equal$

2. *Test statistic*

   $$F = \frac{MSTrt}{MSE}$$

3. *Computation*

A	B	C	
20	21	30	
22	26	24	
25	26	26	
24	27	25	
19	25	30	
110	125	135	370

$n = 5, k = 3 \quad N = 15 \quad y_{1\bullet} = 110 \quad y_{2\bullet} = 125 \quad y_{3\bullet} = 135 \quad y_{\bullet\bullet} = 370$

$$SST = \sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij}^2 - \frac{y_{\bullet\bullet}^2}{N}$$
$$= 20^2 + 22^2 + \cdots + 30^2 - \frac{370^2}{15}$$
$$= 9270 - \frac{370^2}{15}$$
$$= 143.33$$

$$SSTrt = \sum_{i=1}^{k} \frac{y_{i\bullet}^2}{n} - \frac{y_{\bullet\bullet}^2}{N}$$
$$= \frac{110^2}{5} + \frac{125^2}{5} + \frac{135^2}{5} - \frac{370^2}{15}$$
$$= 63.33$$
$$SSE = SST - SSTrt = 143.3 - 63.33 = 80$$

Source	SS	df	MS	F^*
Treatments	63.33	2	31.665	4.75
Error	80	12	6.67	
Total	143.33	14		

4. *Critical region*

reject H_0 if

$$F^* > F_\alpha(k-1, N-k) = f_{0.05}(2,12) = 3.89$$

Note that the level of significance is not given, so we use a standard value of $\alpha = 0.05$

i.e. $F^* > 3.89$

5. *Conclusion*

Reject H_0 at the 0.05 level of significance. The mean times are not the same.

NOTE

1. When the number of observations is the same for all the treatments, we have a balanced design and $N = nk$ as above.
2. When it is not the same then we have an unbalanced design and $N = n_1 + n_2 + \cdots + n_k$ where $n_i = number\ of\ observations\ for\ treatment\ i.$

   The formulae for the sum of squares then become:

$$SST = \sum_{i=1}^{k}\sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{\bullet\bullet})^2$$
$$= \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \frac{y_{\bullet\bullet}^2}{N}$$

$$SSTrt = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2$$
$$= \sum_{i=1}^{k}\frac{y_{i\bullet}^2}{n_i} - \frac{y_{\bullet\bullet}^2}{N}$$
$$SSE = SST - SSTrt$$

EXAMPLE

A production plant manager claimed that there was no difference in the mean times to complete as assembly line job among plants A, B, C and D. Samples from each of the plants yielded the following times (in minutes) to complete the job.

Plant			
A	B	C	D
18	20	23	12
11	14	16	18
14	16	21	17
12	18		13
15			

Test the claim using a 1% level of significance.

SOLUTION: Let

$\mu_1 = mean\ time\ for\ plant\ A$
$\mu_2 = mean\ time\ for\ plant\ B$
$\mu_3 = mean\ time\ for\ plant\ C$
$\mu_4 = mean\ time\ for\ plant\ D$

1. *Testing problem*

   $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$
   $H_1: not\ all\ the\ means\ are\ equal$

2. *Test statistic*

   $$F = \frac{MSTrt}{MSE}$$

3. Computation

A	B	C	D
18	20	23	12
11	14	16	18
14	16	21	17
12	18		13
15			
70	68	60	60

$$k = 4, \quad n_1 = 5 \quad n_2 = 4 \quad n_3 = 3 \quad n_4 = 4 \quad N = 16$$
$$y_1\bullet = 70 \quad y_2\bullet = 68 \quad y_3\bullet = 60 \quad y_4\bullet = 60 \quad y\bullet\bullet = 258$$

$$SSTrt = \sum_{i=1}^{k} \frac{y_{i\bullet}^2}{n_i} - \frac{y_{\bullet\bullet}^2}{N}$$
$$= \frac{70^2}{5} + \frac{68^2}{4} + \frac{60^2}{3} + \frac{60^2}{4} - \frac{258^2}{16}$$
$$= 75.75$$

$$SST = \sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}^2 - \frac{y_{\bullet\bullet}^2}{N}$$
$$= 18^2 + 11^2 + \cdots + 13^2 - \frac{258^2}{16}$$
$$= 4338 - \frac{258^2}{16}$$
$$= 177.75$$
$$SSE = SST - SSTrt = 102$$

Source	SS	df	MS	F^*
Treatments	75.75	3	25.25	2.97
Error	102	12	8.5	
Total	177.75	15		

4. Critical region

reject H_0 if

$$F^* > F_\alpha(k-1, N-1) = f_{0.01}(3,12) = 5.95$$

i.e. $F^* < 5.95$

5. *Conclusion:* Do not reject H_0 at the 0.01 level of significance. The mean times in the 4 plants are not significantly different.

4.4. RANDOMIZED BLOCK DESIGN (RBD)

- ✓ Suppose we want to compare 4 varieties of maize and we have 4 equal plots in different parts of Lusaka, say UNZA, Lilayi, Lusaka West and Kabangwe.
- ✓ How do we perform the comparison?

A. Randomly assign the 4 varieties to the 4 plots and compare the yields (CRD) i.e.

V_1	V_3	V_2	V_4

Problem

- The conditions may be more favourable at one of the locations?
- How do we resolve the problem?

B. Divide the 4 plots into 4 equal subplots and randomly assign each variety a subplot at each of the 4 locations

1	2	3	4
V_2	V_3	V_4	V_1
V_3	V_4	V_1	V_3
V_4	V_1	V_2	V_4
V_1	V_2	V_3	V_2

- ✓ This way we are able to control for the effect of the location of the plot. This is an example of a randomized block design where,
  - each plot in one of the 4 locations is an example of a block and
  - the four varieties of maize are examples of treatments.
- ✓ If we have k treatments and b blocks, then we have the following layout:

		block				
		1	2		b	Total
Treatment	1	y_{11}	y_{12}	$\cdots$	y_{1b}	$y_{1\bullet}$
	2	y_{21}	y_{22}	$\cdots$	y_{2b}	$y_{2\bullet}$
	$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
	k	y_{k1}	y_{k2}	$\cdots$	y_{kb}	$y_{k\bullet}$
Total		$y_{\bullet 1}$	$y_{\bullet 2}$	$\cdots$	$y_{\bullet b}$	$y_{\bullet\bullet}$

Where, y_{ij} is an obervation on the ith treatment and j^{th} block.

NOTE

1. Each treatment appears once in every block.
2. The treatments are allocated at random within each block.
3. The model for this design is

$$Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij} \ , \qquad i = 1,2, \dots, k \ ; j = 1,2, \dots, b$$

Where,

$$Y_{ij} = response\ from\ treatment\ i\ and\ block\ j$$
$$\mu = overall\ mean\ response$$
$$\alpha_i = effect\ of\ i^{th}\ treatment$$
$$\beta_j = effect\ of\ j^{th}\ block$$
$$\varepsilon_{ij} = random\ error$$
$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

(A) TESTING PROBLEM

- ✓ The primary testing problem is finding out whether the treatment means are equal i.e.

$$H_0: \mu_{1\bullet} = \mu_{2\bullet} = \cdots = \mu_{k\bullet} = \mu$$
$$H_1: the\ \mu_{i\bullet}\ are\ not\ all\ equal$$

- ✓ This is equivalent to testing for no treatment effect

$$H_0: \alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$$
$$H_1: at\ least\ one\ \alpha_i \neq 0, i = 1,2, \dots, k$$

- ✓ In addition, it may be desirable to test whether the blocking has been effective i.e.

$$H_0: \beta_1 = \beta_2 = \cdots = \beta_b = 0\ (no\ block\ effect)$$
$$H_1: at\ least\ one\ \beta_j \neq 0, j = 1,2, \dots, b$$

(B) ANOVA TABLE FOR RANDOMIZED BLOCK DESIGN

- ✓ To test the hypothesis

$$H_0: \alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$$
$$H_1: at\ least\ one\ \alpha_i \neq 0, i = 1,2, \dots, k$$

- ✓ we construct the ANOVA table

Source of variation (Source)	Sum of squares (SS)	Degrees of freedom (df)	Mean squares (MS)	F^*
Treatments	$SSTrt$	$k-1$	$MSTrt = \frac{SSTrt}{k-1}$	$\frac{MSTrt}{MSE}$
Block	SSB	$b-1$	$MSB = \frac{SSB}{b-1}$	$\frac{MSB}{MSE}$
Error	SSE	$(k-1)(b-1)$	$MSE = \frac{SSE}{(k-1)(b-1)}$	
Total	SST	$kb-1$		

Where,

$$SSTrt = Treatment\ sum\ of\ squares$$

$$SSTrt = \sum_{i=1}^{k}\sum_{j=1}^{b}(\bar{y}_{i\bullet} - \bar{y}_{\bullet\bullet})^2$$

$$= \sum_{i=1}^{k}\frac{y_{i\bullet}^2}{b} - \frac{y_{\bullet\bullet}^2}{kb}$$

$$SSB = Block\ sum\ of\ squares$$

$$SSTrt = \sum_{i=1}^{k}\sum_{j=1}^{b}\left(\bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet}\right)^2$$

$$= \sum_{i=1}^{b}\frac{y_{\bullet j}^2}{k} - \frac{y_{\bullet\bullet}^2}{kb}$$

$$SST = Total\ sum\ of\ squares$$

$$SST = \sum_{i=1}^{k}\sum_{j=1}^{b}\left(y_{ij} - \bar{y}_{\bullet\bullet}\right)^2$$

$$= \sum_{i=1}^{k}\sum_{j=1}^{b}y_{ij}^2 - \frac{y_{\bullet\bullet}^2}{kb}$$

$$SSE = Error\ sum\ of\ squares$$

$$= SST - SSTrt - SSB$$

We reject $H_0: \alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$ if

$$F^* = \frac{MSTrt}{MSE} > F_\alpha(k-1,(k-1)(b-1))$$

Similarly, we reject $H_0: \beta_1 = \beta_2 = \cdots = \beta_b = 0$ if

$$F^* = \frac{MSB}{MSE} > F_\alpha(b-1,(k-1)(b-1))$$

EXAMPLES

1. The following data represent the final marks obtained by 5 students in Mathematics, English, French and Biology:

Student	Maths	English	French	Biology	Total
1	68	57	73	61	259
2	83	94	91	86	354
3	72	81	63	59	275
4	55	73	77	66	271
5	92	68	75	87	322
Total	370	373	379	359	1481

Perform a randomized block analysis of variance to test the hypothesis that the courses are of equal difficulty.

SOLUTION

In this case the subjects are treatments and the students are blocks.

The model is:

$$Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij} \ , \qquad i = 1,2,3,4 \ ; j = 1,2,3,4,5$$

Where,

$Y_{ij} = mark\ for\ subject\ i\ and\ student\ j$
$\mu = overall\ mean\ mark$
$\alpha_i = effect\ of\ i^{th}\ subject$
$\beta_j = effect\ of\ j^{th}\ student$
$\varepsilon_{ij} = random\ error$
$\varepsilon_{ij} \sim N(0,\sigma^2)$

1. Testing problem

$H_0: \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$
$H_1: at\ leat\ one\ \alpha_i \neq 0, i = 1,2,3,4$

2. *Test statistic*

$$F = \frac{MSTrt}{MSE}$$

3. *Computation*

$Treatments$:

$y_{1\bullet} = 370 \quad y_{2\bullet} = 373 \quad y_{3\bullet} = 379 \quad y_{4\bullet} = 359 \quad y_{\bullet\bullet} = 1481 \quad b = 5$

$$\begin{aligned} SSTrt &= \sum_{i=1}^{k} \frac{y_{i\bullet}^2}{b} - \frac{y_{\bullet\bullet}^2}{kb} \\ &= \frac{370^2}{5} + \frac{373^2}{5} + \frac{379^2}{5} + \frac{359^2}{5} - \frac{1481^2}{20} \\ &= 42.15 \end{aligned}$$

$Blocks$:

$y_{\bullet 1} = 259 \quad y_{\bullet 2} = 354 \quad y_{\bullet 3} = 275 \quad y_{\bullet 4} = 271 \quad y_{\bullet 5} = 322 \quad k = 4$

$$\begin{aligned} SSB &= \sum_{i=1}^{b} \frac{y_{\bullet j}^2}{k} - \frac{y_{\bullet\bullet}^2}{kb} \\ &= \frac{259^2}{4} + \frac{354^2}{4} + \frac{275^2}{4} + \frac{271^2}{4} + \frac{322^2}{4} - \frac{1481^2}{20} \\ &= 1618.7 \end{aligned}$$

$$\begin{aligned} SST &= \sum_{i=1}^{k}\sum_{j=1}^{b} y_{ij}^2 - \frac{y_{\bullet\bullet}^2}{kb} \\ &= 68^2 + 83^2 + \cdots + 87^2 - \frac{1481^2}{20} \\ &= 112441 - \frac{1481^2}{20} \\ &= 2772.95 \end{aligned}$$

$SSE = SST - SSTrt - SSB = 2772.95 - 42.15 - 1618.7 = 1112.1$

Source	SS	Df	MS	F^*
Treatments	42.15	3	14.05	0.15
Block	1618.7	4	404.675	4.37
Error	1112.1	12	92.675	
Total	2772.95	19		

4. *Critical region*

   reject H_0 if

   $$F^* > F_\alpha(k-1,(k-1)(b-1)) = f_{0.05}(3,12) = 3.49$$

   i.e. $F^* < 3.49$

5. *Conclusion*

   Do not reject H_0 at the 0.05 level of significance. The courses are of equal difficulty.

   NOTE:

   We would reject the hypothesis that the students have equal ability since

   $$F^* = 4.37 > f_{0.05}(4,12) = 3.26$$

2. Three varieties of potatoes are being compared for yield. The experiment was conducted by assigning each variety at random to 3 equal-size plots at each of 4 different locations. The following yields for varieties A, B and C in 100 kg per plot, were recorded:

Location 1	***Location 2***	***Location 3***	***Location 4***
B 13	C 21	C 9	A 11
A 18	A 20	B 12	C 10
C 12	B 23	A 14	B 17

Conduct an analysis of variance using the randomized complete block model to test the hypothesis that there is no difference in the yields of the 3 varieties of potatoes.

SOLUTION

Treatments = varieties

Blocks = locations

Model:

$$Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}\ , \qquad i = 1,2,3\ ; j = 1,2,3,4$$

Where,

$$Y_{ij} = yield\ from\ i^{th}\ variety\ and\ j^{th}\ location$$
$$\mu = overall\ mean\ yield$$
$$\alpha_i = effect\ of\ i^{th}\ variety$$
$$\beta_j = effect\ of\ j^{th}\ location$$
$$\varepsilon_{ij} = random\ error$$
$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

1. *Testing problem*

   $H_0: \alpha_1 = \alpha_2 = \alpha_3 = 0$
   $H_1: not\ all\ \alpha_i's\ are\ equal\ to\ zero$

2. *Test statistic*

   $$F = \frac{MSTrt}{MSE}$$

3. *Computation*

   $Treatments$:
   $y_{1\bullet} = 18 + 20 + 14 + 11 = 63$
   $y_{2\bullet} = 13 + 23 + 12 + 17 = 65$
   $y_{3\bullet} = 12 + 21 + 9 + 10 = 52$
   $y_{\bullet\bullet} = 180$

$$\begin{aligned} SSTrt &= \sum_{i=1}^{k} \frac{y_{i\bullet}^2}{b} - \frac{y_{\bullet\bullet}^2}{kb} \\ &= \frac{63^2}{4} + \frac{65^2}{4} + \frac{52^2}{4} - \frac{180^2}{12} \\ &= 24.5 \end{aligned}$$

   $Blocks$:
   $y_{\bullet 1} = 13 + 18 + 12 = 43$
   $y_{\bullet 2} = 21 + 20 + 23 = 64$
   $y_{\bullet 3} = 9 + 12 + 14 = 35$
   $y_{\bullet 4} = 11 + 10 + 17 = 38$

$$\begin{aligned} SSB &= \sum_{i=1}^{b} \frac{y_{\bullet j}^2}{k} - \frac{y_{\bullet\bullet}^2}{kb} \\ &= \frac{43^2}{3} + \frac{64^2}{3} + \frac{35^2}{3} + \frac{38^2}{3} - \frac{180^2}{12} \\ &= 171.33 \end{aligned}$$

$$SST = \sum_{i=1}^{k}\sum_{j=1}^{b} y_{ij}^2 - \frac{y_{\bullet\bullet}^2}{kb}$$

$$= 13^2 + 18^2 + \cdots + 17^2 - \frac{180^2}{12}$$

$$= 2938 - \frac{180^2}{12}$$

$$= 238$$

$$SSE = SST - SSTrt - SSB = 42.17$$

Source	SS	df	MS	F^*
Treatments	24.5	2	12.25	1.74
Block	171.33	3	57.11	8.13
Error	42.17	6	7.028	
Total	238	11		

4. *Critical region*

   reject H_0 if

   $$F^* > F_\alpha(k-1,(k-1)(b-1)) = f_{0.05}(2,6) = 5.14$$

   i.e. $F^* < 5.14$

5. *Conclusion:* Do not reject H_0 at the 0.05 level of significance. There is no significant difference in the yields of the three varieties.